

Chapter VI

Fractal Interpolation

1 Introduction: Applications for Fractal Functions

Euclidean geometry, trigonometry, and calculus have taught us to think about modelling the shapes we see in the real world in terms of straight lines, circles, parabolas, and other simple curves. Consequences of this way of thinking are abundant in our everyday lives. They include the design of household objects; the common usage of drafting tables, straight-edges, and compasses; and the “applications” that accompany introductory calculus courses. We note in particular the provision of functions for drawing points, lines, polygons, and circles in computer graphics software such as MacPaint and Turbobasic. Most computer graphics hardware is designed specifically to provide rapid computation and display of classical geometrical shapes.

Euclidean geometry and elementary functions, such as sine, cosine, and polynomials, are the basis of the traditional method for analyzing experimental data. Consider an experiment that measures values of a real-valued function $F(x)$ as a function of a real variable x . For example, $F(x)$ may denote a voltage as a function of time, as in the experiments on the jet-engine exhaust described in Example 3.3 in Chapter V. The experiment may be a numerical experiment on a computer. In any case the result of the experiment will be a collection of data of the form:

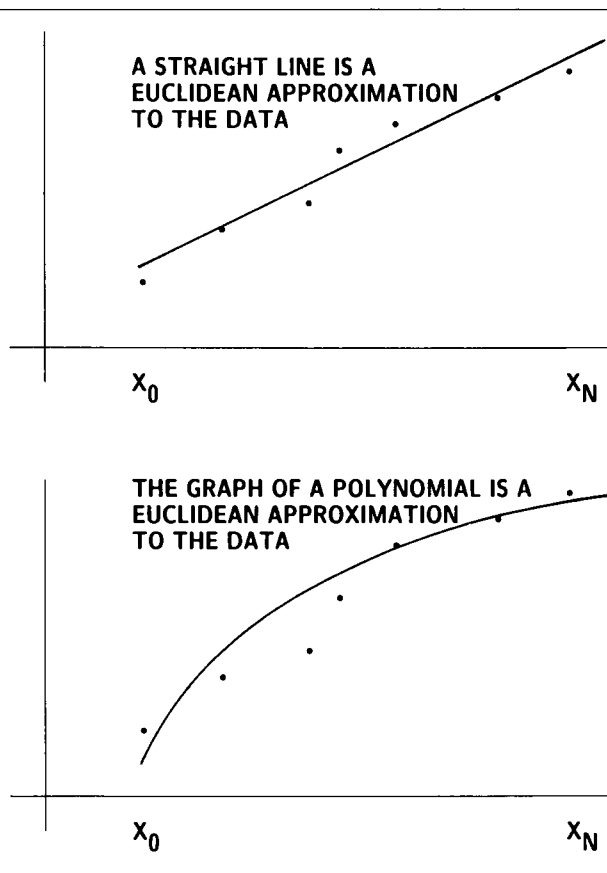
$$\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}.$$

Here N is a positive integer, $F_i = F(x_i)$, and the x_i 's are real numbers such that

$$x_0 < x_1 < x_2 < x_3 < \dots < x_N.$$

The traditional method for analyzing this data begins by representing it graphically as a subset of $\mathbb{R}^2$. That is, the data points are plotted on graph paper. Next the graphical data is analyzed geometrically. For example, one may seek a straight line segment that is a good approximation to the graph of the data. Or else, one might construct a polynomial of as low degree as possible, whose graph is a good fit to the data over the interval $[x_0, x_N]$. In place of a polynomial, a linear combination of elementary functions might be used. The goal is always the same: to represent the

Figure VI.159. Illustration of the process whereby experimental data is represented graphically and modelled geometrically by means of a classical geometrical entity, such as a straight line or a polynomial fit to the data.



data, viewed as a subset of $\mathbb{R}^2$, by a classical geometrical entity. This entity is represented by a simple formula, one that can be communicated easily to someone else. The process is illustrated in Figure VI.159.

Elementary functions, such as trigonometric functions and rational functions, have their roots in Euclidean geometry. They share the feature that when their graphs are “magnified” sufficiently, locally they “look like” straight lines. That is, the tangent line approximation can be used effectively in the vicinity of most points. Moreover, the fractal dimension of the graphs of these functions is always 1. These elementary “Euclidean” functions are useful not only because of their geometrical content but because they can be expressed by simple formulas. We can use them to pass information easily from one person to another. They provide a common language for our scientific work. Moreover, elementary functions are used extensively in scientific computation, computer-aided design, and data analysis because they can be stored in small files and computed by fast algorithms.

Graphics systems founded on traditional geometry are effective for making pictures of man-made objects, such as bricks, wheels, roads, buildings, and cogs. This

is not surprising, since these objects were designed in the first place using Euclidean geometry. However, it is desirable for graphics systems to be able to deal with a wider range of problems.

In this chapter we introduce fractal interpolation functions. The graphs of these functions can be used to approximate image components such as the profiles of mountain ranges, the tops of clouds, stalactite- hung roofs of caves, and horizons over forests, as illustrated in Figure VI.160. Rather than treating the image component as arising from a random assemblage of objects, such as individual mountains, cloudlets, stalactites, or tree tops, one models the image component as an interrelated single system. Such components are not well described by elementary functions or Euclidean graphics functions.

Fractal interpolation functions also provide a new means for fitting experimental data. Clearly it does not suffice to make a polynomial “least-squares” fit to the wild experimental data of Strahle for the temperature in a jet exhaust as a function of time, as illustrated in Figure V.147. Nor would classical geometry be a good tool for the analysis of voltages at a point in the human brain as read by an electroencephalograph. However, fractal interpolation functions can be used to “fit” such experimental data: that is, the graph of the fractal interpolation function can be made close, in the Hausdorff metric, to the data. Moreover, one can ensure that the fractal dimension of the graph of the fractal interpolation function agrees with that of the data, over an appropriate range of scales. This idea is illustrated in Figure VI.161.

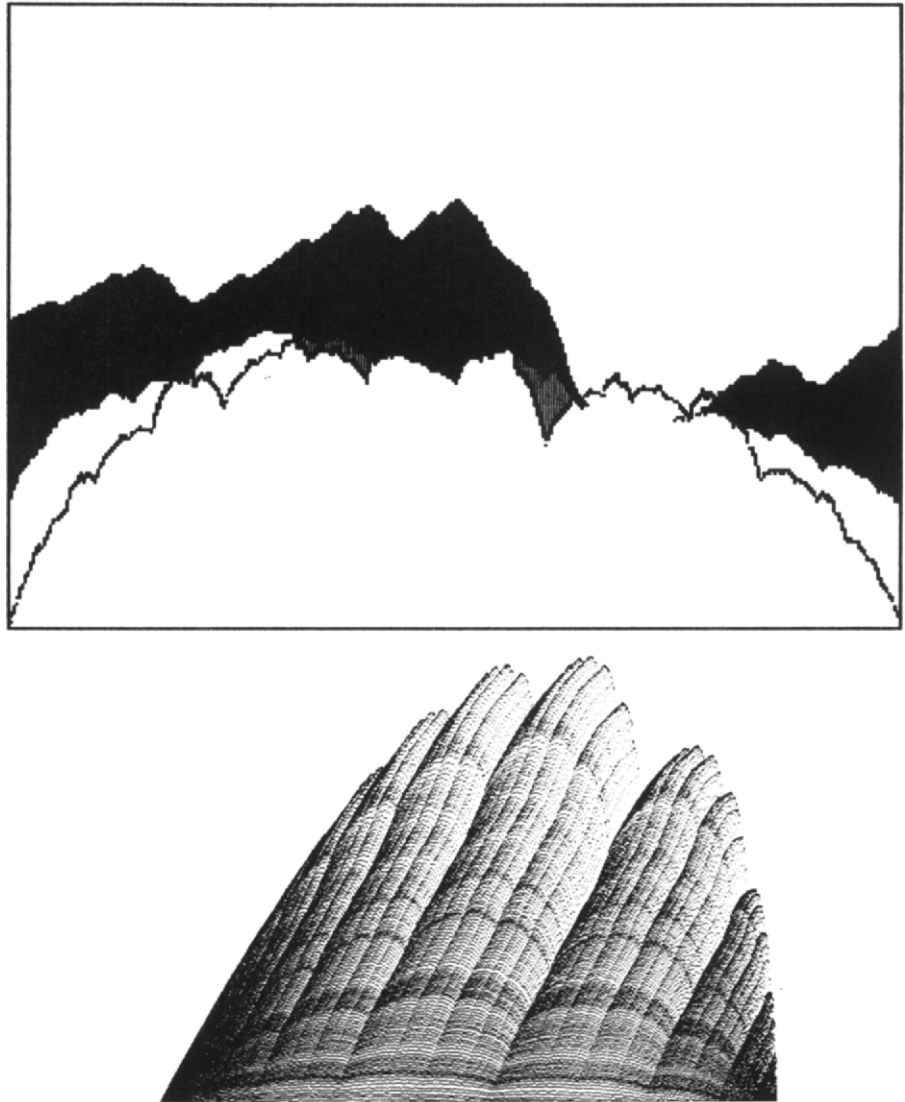
Fractal interpolation functions share with elementary functions that they are of a geometrical character, that they can be represented succinctly by “formulas,” and that they can be computed rapidly. The main difference is their fractal character. For example, they can have a noninteger fractal dimension. They are easy to work with—once one is accustomed to working with sets rather than points and with IFS theory using affine maps. If we start to pass them from one to another, fractal functions will become part of the common language of science. So read on!

Examples & Exercises

1.1. Write an essay on the influences of Euclidean geometry on the way in which we view the physical world. How does fractal geometry change that view?

1.2. Find the linear approximation $l(x)$ to the function $f(x) = \sin(x)$, about the point $x = 0$. Let $\epsilon > 0$. Find the linear change of coordinates $(x', y') = \theta(x, y)$ in $\mathbb{R}^2$, such that $\theta([0, \epsilon] \times [0, \epsilon]) = [0, 1] \times [0, 1]$. Let $l'(x')$ denote the function $l(x)$ represented in the new coordinate system. Let $f'(x')$ denote the function $f(x)$ in the new coordinate system. Let L denote the graph of $l'(x')$ for $x' \in [0, 1]$ and let G denote the graph of $f'(x')$ for $x' \in [0, 1]$. How small must ϵ be chosen to ensure that the Hausdorff distance from L to G is less than 0.01? The Hausdorff distance should be computed with respect to the Manhattan metric in $\mathbb{R}^2$.

Figure VI.160. The fractal interpolation functions introduced in this chapter may be used in computer graphics software packages to provide a simple means for rendering profiles of mountain ranges, the tops of clouds, and horizons over forests.



2 Fractal Interpolation Functions

Definition 2.1 A set of data is a set of points of the form $\{(x_i, F_i) \in \mathbb{R}^2 : i = 0, 1, 2, \dots, N\}$, where

$$x_0 < x_1 < x_2 < x_3 < \dots < x_N.$$

An interpolation function corresponding to this set of data is a continuous function $f : [x_0, x_N] \rightarrow \mathbb{R}$ such that

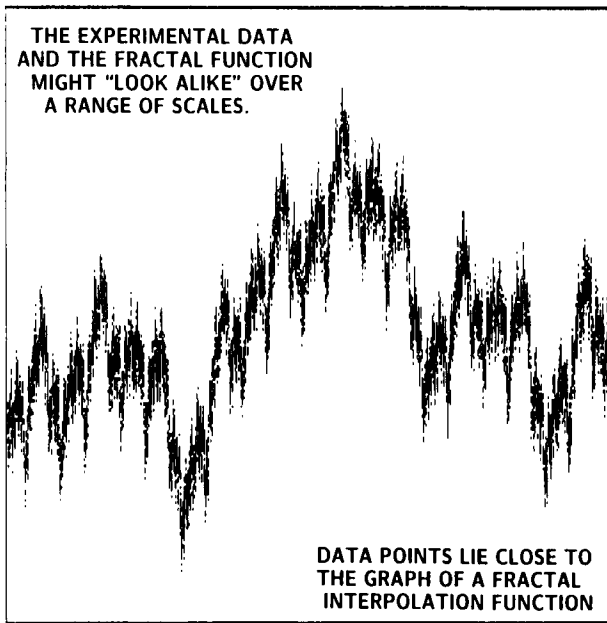


Figure VI.161. This figure illustrates the idea of using a fractal interpolation function to fit experimental data. The graph of the interpolation function may be close, in the Hausdorff metric, to the graph of the experimental data. The fractal dimension of the interpolation function may agree with that of the data over an appropriate range of scales.

$$f(x_i) = F_i \quad \text{for } i = 1, 2, \dots, N.$$

The points $(x_i, F_i) \in \mathbb{R}^2$ are called the interpolation points. We say that the function f interpolates the data and that (the graph of) f passes through the interpolation points.

Examples & Exercises

2.1. The function $f(x) = 1 + x$ is an interpolation function for the set of data

$$\{(0, 1), (1, 2)\}.$$

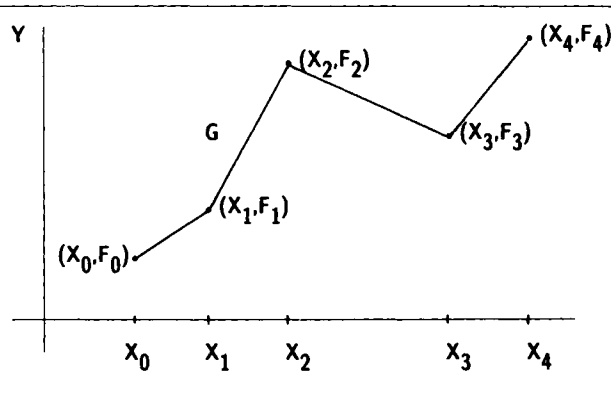
Consider the hyperbolic IFS $\{\mathbb{R}^2; w_1, w_2\}$, where

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ 0.5 \end{pmatrix}, \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0.5 \\ 1 \end{pmatrix}. \end{aligned}$$

Let G denote the attractor of the IFS. Then it is readily verified that G is the straight line segment that connects the pair of points $(0, 1)$ and $(1, 2)$. In other words, G is the graph of the interpolation function $f(x)$ over the interval $[0, 1]$.

2.2. Let $\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$ denote a set of data. Let $f : [x_0, x_N] \rightarrow \mathbb{R}$ denote the unique continuous function that passes through the interpolation points and is linear on each of the subintervals $[x_{i-1}, x_i]$. That is,

Figure VI.162. Graph of the piecewise linear interpolation function $f(x)$ through the interpolation points $\{(F_i, x_i) : i = 0, 1, 2, 3, 4\}$. This graph is also the attractor of an IFS of the form $\{\mathbb{R}^2; w_n, n = 1, 2, 3, 4\}$, where the maps are affine.



$$f(x) = F_{i-1} + \frac{(x - x_{i-1})}{(x_i - x_{i-1})}(F_i - F_{i-1}) \text{ for } x \in [x_{i-1}, x_i], i = 1, 2, \dots, N.$$

The function $f(x)$ is called a piecewise linear interpolation function. The graph of $f(x)$ is illustrated in Figure VI.162. This graph, G , is also the attractor of an IFS of the form $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$, where the maps are affine. In fact,

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix},$$

where

$$\begin{aligned} a_n &= \frac{(x_n - x_{n-1})}{(x_N - x_0)}, & e_n &= \frac{(x_N x_{n-1} - x_0 x_n)}{(x_N - x_0)}, \\ c_n &= \frac{(F_n - F_{n-1})}{(x_N - x_0)}, & f_n &= \frac{(x_N F_{n-1} - x_0 F_n)}{(x_N - x_0)}, \text{ for } n = 1, 2, \dots, N. \end{aligned}$$

Notice that the IFS may not be hyperbolic with respect to the Euclidean metric in $\mathbb{R}^2$. Can you prove that, nonetheless, G is the unique nonempty compact subset of $\mathbb{R}^2$ such that

$$G = \bigcup_{n=1}^N w_n(G)?$$

2.3. Verify the claims in exercise 2.2 in the case of the data set $\{(0, 0), (1, 3), (2, 0)\}$ by applying either the Deterministic Algorithm, Program 1, Chapter III, or the Random Iteration Algorithm, Program 2 of Chapter III. You will need to modify the programs slightly.

2.4. The parabola defined by $f(x) = 2x - x^2$ on the interval $[0, 2]$ is an interpolation function for the set of data $\{(0, 0), (1, 1), (2, 0)\}$. Let G denote the graph of $f(x)$. That is

$$G = \{(x, 2x - x^2) : x \in [0, 2]\}.$$

Then we claim that G is the attractor of the hyperbolic IFS $\{\mathbb{R}^2; w_1, w_2\}$, where

$$\begin{aligned} w_1 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0.5 & 0 \\ 0.5 & 0.25 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}, \\ w_2 \begin{pmatrix} x \\ y \end{pmatrix} &= \begin{pmatrix} 0.5 & 0 \\ -0.5 & 0.25 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \end{aligned}$$

We verify this claim directly. We simply note that for all $x \in [0, 2]$,

$$\begin{aligned} w_1 \begin{pmatrix} x \\ f(x) \end{pmatrix} &= \begin{pmatrix} \frac{1}{2}x \\ 2(\frac{1}{2}x) - (\frac{1}{2}x)^2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x \\ f(\frac{1}{2}x) \end{pmatrix}, \\ w_2 \begin{pmatrix} x \\ f(x) \end{pmatrix} &= \begin{pmatrix} 1 + \frac{1}{2}x \\ 2(1 + \frac{1}{2}x) - (1 + \frac{1}{2}x)^2 \end{pmatrix} = \begin{pmatrix} 1 + \frac{1}{2}x \\ f(1 + \frac{1}{2}x) \end{pmatrix}. \end{aligned}$$

As x varies over $[0, 2]$, the right-hand side of the first equation yields the part of the graph of $f(x)$ lying over the interval $[0, 1]$, while the right-hand side of the second equation yields the part of the graph of $f(x)$ lying over the interval $[1, 2]$. Hence $G = w_1(G) \cup w_2(G)$. Since $G \in \mathcal{H}(\mathbb{R}^2)$ we conclude that it is the attractor of the IFS. Notice that the IFS is just-touching.

2.5. Find a hyperbolic IFS of the form $\{\mathbb{R}^2; w_1, w_2\}$, where w_1 and w_2 are affine transformations in $\mathbb{R}^2$, whose attractor is the graph of the quadratic function that interpolates the data $\{(0, 0), (1, 1), (2, 4)\}$.

Let a set of data $\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$ be given. We explain how one can construct an IFS in $\mathbb{R}^2$ such that its attractor, which we denote by G , is the graph of a continuous function $f : [x_0, x_N] \rightarrow \mathbb{R}$, which interpolates the data. Throughout we will restrict our attention to affine transformations. The usage of more general transformations is discussed in [Barnsley 1988f].

We consider an IFS of the form $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$, where the maps are affine transformations of the special structure

$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix}.$$

The transformations are constrained by the data according to

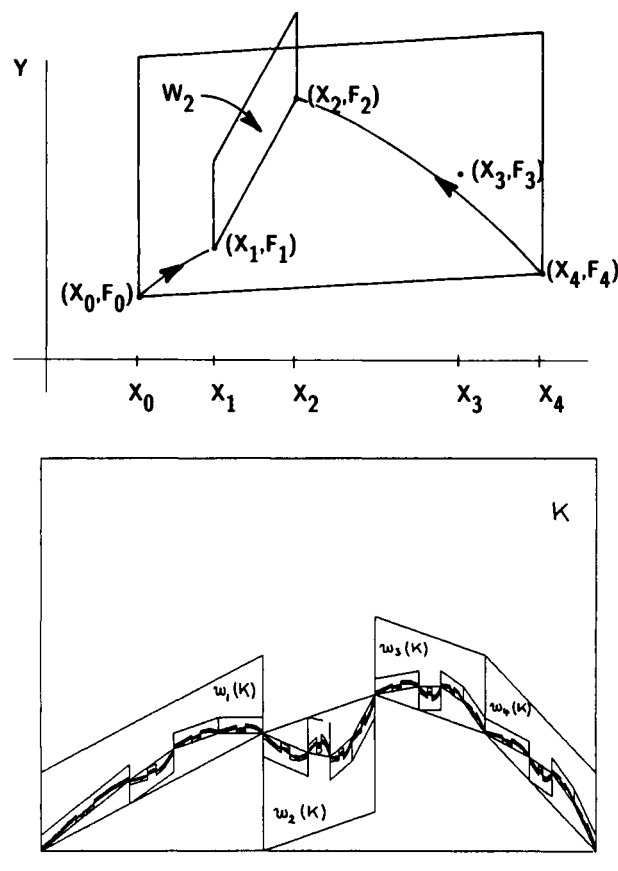
$$w_n \begin{pmatrix} x_0 \\ F_0 \end{pmatrix} = \begin{pmatrix} x_{n-1} \\ F_{n-1} \end{pmatrix} \quad \text{and} \quad w_n \begin{pmatrix} x_N \\ F_N \end{pmatrix} = \begin{pmatrix} x_n \\ F_n \end{pmatrix} \quad \text{for } n = 1, 2, \dots, N.$$

The situation is summarized in Figure VI.163.

Let $n \in \{1, 2, 3, \dots, N\}$. The transformation w_n is specified by the five real numbers a_n, b_n, c_n, d_n , and e_n , which must obey the four linear equations

$$\begin{aligned} a_n x_0 + e_n &= x_{n-1}, \\ a_n x_N + e_n &= x_n, \\ c_n x_0 + d_n F_0 + f_n &= F_{n-1}, \\ c_n x_N + d_n F_N + f_n &= F_n. \end{aligned}$$

Figure VI.163. Two illustrations showing how an IFS of shear transformations is used to construct a fractal interpolation function. (Produced by Peter Massopust.)



It follows that there is effectively one free parameter in each transformation. We choose this parameter to be d_n for the following reason. The transformation w_n is a *shear* transformation: it maps lines parallel to the y -axis into lines parallel to the y -axis. Let L denote a line segment parallel to the y -axis. Then $w_n(L)$ is also a line segment parallel to the y -axis. The ratio of the length of $w_n(L)$ to the length of L is $|d_n|$. We call d_n the *vertical scaling factor* in the transformation w_n . By choosing d_n to be the free parameter, we are able to specify the vertical scaling produced by the transformation. With $d_n = 0$, $n = 1, 2, \dots, N$, one recovers the piecewise linear interpolation function. In section 6.2 we will show that these parameters determine the fractal dimension of the attractor of the IFS.

Let d_n be any real number. We demonstrate that we can always solve the above equations for a_n , c_n , e_n , and f_n in terms of the data and d_n . We find

$$a_n = \frac{(x_n - x_{n-1})}{(x_N - x_0)}, \quad (1)$$

$$e_n = \frac{(x_N x_{n-1} - x_0 x_n)}{(x_N - x_0)}, \quad (2)$$

$$c_n = \frac{(F_n - F_{n-1})}{(x_N - x_0)} - \frac{d_n(F_N - F_0)}{(x_N - x_0)}, \quad (3)$$

$$f_n = \frac{(x_N F_{n-1} - x_0 F_n)}{(x_N - x_0)} - d_n \frac{(x_N F_0 - x_0 F_N)}{(x_N - x_0)}. \quad (4)$$

Now let $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$ denote the IFS defined above. Let the vertical scaling factor d_n obey $0 \leq d_n < 1$ for $n = 1, 2, \dots, N$. Even with this condition, the IFS is not in general hyperbolic on the metric space $(\mathbb{R}^2, \text{Euclidean})$. Despite this, let us see what happens if we apply the Random Iteration Algorithm to the IFS.

Here we present Program 2 in Chapter III modified so that the input data consists of the interpolation points and vertical scaling factors. It is written for $N = 3$ and the data set

$$\{(0, 0), (30, 50), (60, 40), (100, 10)\}.$$

The vertical scaling factors are input by the user during execution of the code. The program calculates the coefficients in the shear transformations from the data, and then applies the Random Iteration Algorithm to the resulting IFS. The program is written in BASIC. It runs without modification on an IBM PC with enhanced graphics adaptor and Turbobasic. On any line the words preceded by a ' are comments: they are not part of the program.

Program 1.

```
x[0] =0 : x[1] =30 : x[2] =60 : x[3] =100 ' Data set

F[0] =0 : F[1] =50 : F[2] =40 : F[3] =10

'Vertical Scaling Factors
input "enter scaling factors d(1), d(2), and d(3)",
    d(1),d(2),d(3)

'Calculate the shear transformations from the Data
'and Vertical Scaling Factors
for n =1 to 3

b = x[3]-x[0] : a[n] = (x[n]-x[n-1])/b
```

```

e[n] = (x[3]*x[n-1]-x[0]*x[n])/b

c[n] = (F[n]-F[n-1]-d[n]*(F[3]-F[0]))/b

ff[n] = (x[3]*F[n-1]-x[0]*F[n]-d[n]*(x[3]*F[0]-x[0]*F[3]))/b

next

screen 2 : cls ' initialize graphics

window(0,0)-(100,100) 'change this to zoom and/or pan

x =0: y =0 'initial point from which the random
           'iteration begins

for n =1 to 1000 'Random Iteration Algorithm

k =int(3*rnd-0.0001)+1

newx = a[k]*x + e[k]

newy = c[k]*x + d[k]*y + ff[k]

x =newx : y =newy

pset(x,y) 'plot the most recently computed point on the screen

next

end

```

The result of running an adaptation of this program on a Masscomp workstation and then printing the contents of the graphics screen is presented in Figure VI.164. In this case $d_1 = 0.5$, $d_2 = -0.5$, and $d_3 = 0.23$. Notice that if the size of the plotting window is decreased, for example by replacing the window call by WINDOW (0,0)-(50,50), then a portion of the image is plotted at a higher resolution. The number of iterations can be increased to improve the quality of the computed image.

Examples & Exercises

2.6. Rewrite Program 1 in a form suitable for your own computer environment, then run it and obtain hardcopy of the output.

2.7. Vary the data used by Program 1. Verify, by means of computergraphical experiments, that the corresponding IFS always seems to have a unique attractor, provided that the vertical scaling factors are less than 1 in norm. Verify that, provided suffi-

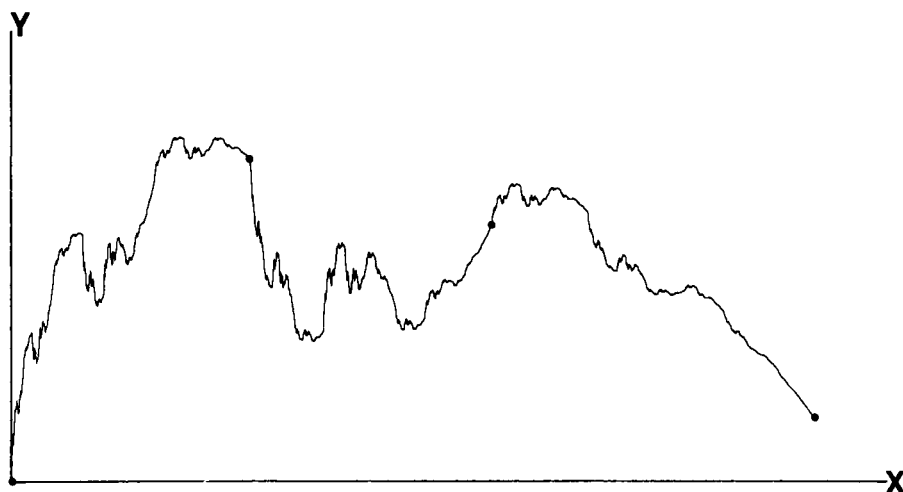


Figure VI.164. The result of running Program 1 with vertical scaling factors 0.5, -0.5 , and 0.23 . It appears that the corresponding IFS possesses a unique attractor that is the graph of a function that passes through the interpolation points $\{(0, 0), (30, 50), (60, 40), (100, 10)\}$. Is there a metric such that the IFS is hyperbolic?

ciently many points are plotted, the attractor always contains the data points, and that it looks like the graph of a function.

2.8. Show that the shear transformations w_n , described above, need not be contractions in the Euclidean metric, even though the magnitudes of the vertical scaling factors are less than 1. Once you have found such an example, use Program 1, suitably modified, to obtain graphical evidence concerning the possible existence of an attractor. You are supposed to discover that even though the IFS is not hyperbolic in the Euclidean metric, it appears to possess an attractor.

2.9. Use Program 1 to verify that the attractor of the IFS in exercise 2.4 is a parabola.

We now give the theoretical basis for our experimental observations.

Theorem 2.1 *Let N be a positive integer greater than 1. Let $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$ denote the IFS defined above, associated with the data set*

$$\{(x_n, F_n) : n = 1, 2, \dots, N\}.$$

Let the vertical scaling factor d_n obey $0 \leq d_n < 1$ for $n = 1, 2, \dots, N$. Then there is a metric d on $\mathbb{R}^2$, equivalent to the Euclidean metric, such that the IFS is hyperbolic with respect to d . In particular, there is a unique nonempty compact set $G \subset \mathbb{R}^2$ such that

$$G = \cup_{n=1}^N w_n(G).$$

Proof We define a metric d on $\mathbb{R}^2$ by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + \theta |y_1 - y_2|,$$

where θ is a positive real number, which we specify below. We leave it as an exercise to the reader to prove that this metric is equivalent to the Euclidean metric on $\mathbb{R}^2$. Let $n \in \{1, 2, \dots, N\}$. Let the numbers a_n, c_n, e_n, f_n , be defined by equations 1, 2, 3, and 4. Then we have

$$\begin{aligned} d(w_n(x_1, y_1), w_n(x_2, y_2)) &= d((a_n x_1 + e_n, c_n x_1 + d_n y_1 + f_n), (a_n x_2 + e_n, c_n x_2 + d_n y_2 + f_n)) \\ &= a_n |x_1 - x_2| + \theta |c_n(x_1 - x_2) + d_n(y_1 - y_2)| \\ &\leq (|a_n| + \theta |c_n|) |x_1 - x_2| + \theta |d_n| |y_1 - y_2|. \end{aligned}$$

Now notice that $|a_n| = |x_n - x_{n-1}|/|x_N - x_0| < 1$ because $N \geq 2$. If $c_1 = c_2 \cdots = c_n = 0$, then we choose $\theta = 1$. Otherwise we choose

$$\theta = \frac{\min\{(2 - |a_n|) : n = 1, 2, \dots, N\}}{\max\{2|c_n| : n = 1, 2, \dots, N\}}.$$

Then it follows that

$$\begin{aligned} d(w_n(x_1, y_1), w_n(x_2, y_2)) &\leq (|a_n| + \theta |c_n|) |x_1 - x_2| + \theta |d_n| |y_1 - y_2| \\ &\leq a |x_1 - x_2| + \theta \delta |y_1 - y_2| \\ &\leq \max\{a, \delta\} d((x_1, y_1), (x_2, y_2)), \end{aligned}$$

where

$$\begin{aligned} a &= (1 + a_n - \frac{\max\{|a_n| : n = 1, 2, \dots, N\}}{2}) < 1, \\ \delta &= \max\{|d_n| : n = 1, 2, \dots, N\} < 1. \end{aligned}$$

This completes the proof.

Theorem 2.2 *Let N be a positive integer greater than 1. Let $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$ denote the IFS defined above, associated with the data set $\{(x_n, F_n) : n = 1, 2, \dots, N\}$. Let the vertical scaling factor d_n obey $0 \leq d_n < 1$ for $n = 1, 2, \dots, N$, so that the IFS is hyperbolic. Let G denote the attractor of the IFS. Then G is the graph of a continuous function $f : [x_0, x_N] \rightarrow \mathbb{R}$, which interpolates the data $\{(x_i, F_i) : i = 1, 2, \dots, N\}$. That is,*

$$G = \{(x, f(x)) : x \in [x_0, x_N]\},$$

where

$$f(x_i) = F_i \quad \text{for } i = 0, 1, 2, 3, \dots, N.$$

Proof Let $\mathcal{F}$ denote the set of continuous functions $f : [x_0, x_N] \rightarrow \mathbb{R}$ such that $f(x_0) = F_0$ and $f(x_N) = F_N$. We define a metric d on $\mathcal{F}$ by

$$d(f, g) = \max\{|f(x) - g(x)| : x \in [x_0, x_N]\} \quad \text{for all } f, g \text{ in } \mathcal{F}.$$

Then $(\mathcal{F}, d)$ is a complete metric space—see, for example [Rudin 1966], or prove it yourself.

Let the real numbers a_n, c_n, e_n, f_n , be defined by equations 1, 2, 3, and 4. Define a mapping $T : \mathcal{F} \rightarrow \mathcal{F}$ by

$$(Tf)(x) = c_n l_n^{-1}(x) + d_n f(l_n^{-1}(x)) + f_n \text{ for } x \in [x_{n-1}, x_n], \text{ for } n = 1, 2, \dots, N,$$

where $l_n : [x_0, x_N] \rightarrow [x_{n-1}, x_n]$ is the invertible transformation

$$l_n(x) = a_n x + e_n.$$

We verify that T does indeed take $\mathcal{F}$ into itself. Let $f \in \mathcal{F}$. Then the function $(Tf)(x)$ obeys the endpoint conditions because

$$\begin{aligned} (Tf)(x_0) &= c_1 l_1^{-1}(x_0) + d_1 f(l_1^{-1}(x_0)) + f_1 \\ &= c_1 x_0 + d_1 f(x_0) + f_1 = c_1 x_0 + d_1 F_0 + f_1 \\ &= F_0 \end{aligned}$$

and

$$\begin{aligned} (Tf)(x_N) &= c_N l_N^{-1}(x_N) + d_N f(l_N^{-1}(x_N)) + f_N \\ &= c_N x_N + d_N f(x_N) + f_N \\ &= c_N x_N + d_N F_N. \end{aligned}$$

The reader can prove that $(Tf)(x)$ is continuous on the interval $[x_{n-1}, x_n]$ for $n = 1, 2, \dots, N$. Then it remains to demonstrate that $(Tf)(x)$ is continuous at each of the points $x_1, x_2, x_3, \dots, x_{N-1}$. At each of these points the value of $(Tf)(x)$ is apparently defined in two different ways. For $n \in \{1, 2, \dots, N-1\}$ we have

$$\begin{aligned} (Tf)(x_n) &= c_{n+1} l_{n+1}^{-1}(x_n) + d_{n+1} f(l_{n+1}^{-1}(x_n)) + f_{n+1} \\ &= c_{n+1} x_0 + d_{n+1} f(x_0) + f_{n+1} = F_n \end{aligned}$$

and also

$$\begin{aligned} (Tf)(x_n) &= c_n l_n^{-1}(x_n) + d_n f(l_n^{-1}(x_n)) + f_n \\ &= c_n x_N + d_n f(x_N) + f_n = F_n, \end{aligned}$$

so both methods of evaluation lead to the same result. We conclude that T does indeed take $\mathcal{F}$ into $\mathcal{F}$.

Now we show that T is a contraction mapping on the metric space $(\mathcal{F}, d)$. Let $f, g \in \mathcal{F}$. Let $n \in \{1, 2, \dots, N\}$ and $x \in [x_{n-1}, x_n]$. Then

$$|(Tf)(x) - (Tg)(x)| = |d_n| |f(l_n^{-1}(x)) - g(l_n^{-1}(x))| \leq |d_n| d(f, g).$$

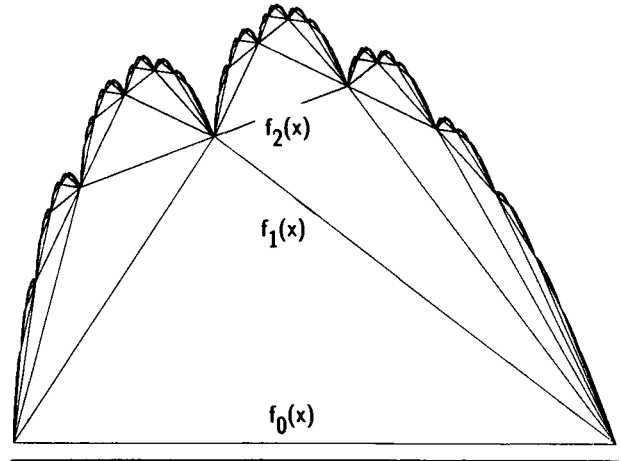
It follows that

$$d(Tf, Tg) \leq \delta d(f, g) \quad \text{where } \delta = \max\{|d_n| : n = 1, 2, \dots, N\} < 1.$$

We conclude that $T : \mathcal{F} \rightarrow \mathcal{F}$ is a contraction mapping. The Contraction Mapping Theorem implies that T possesses a unique fixed point in $\mathcal{F}$. That is, there exists a function $f \in \mathcal{F}$ such that

$$(Tf)(x) = f(x) \quad \text{for all } x \in [x_0, x_N].$$

Figure VI.165. A sequence of functions $\{f_{n+1}(x) = (Tf_n)(x)\}$ converging to the fixed point of the mapping $T : \mathcal{F} \rightarrow \mathcal{F}$ used in the proof of Theorem 2.2. This is another example of a contraction mapping doing its work.



The reader should convince himself that f passes through the interpolation points.

Let $\tilde{G}$ denote the graph of f . Notice that the equations that define T can be rewritten

$$(Tf)(a_n x + e_n) = c_n x + d_n f(x) + f_n \text{ for } x \in [x_0, x_N], \text{ for } n = 1, 2, \dots, N,$$

which implies that

$$\tilde{G} = \cup_{n=1}^N w_n(\tilde{G}).$$

But $\tilde{G}$ is a nonempty compact subset of $\mathbb{R}^2$. By Theorem 2.1 there is only one nonempty compact set G , the attractor of the IFS, which obeys the latter equation. It follows that $G = \tilde{G}$. This completes the proof.

Definition 2.2 *The function $f(x)$ whose graph is the attractor of an IFS as described in Theorems 2.1 and 2.2, above, is called a fractal interpolation function corresponding to the data $\{(x_i, F_i) : i = 1, 2, \dots, N\}$.*

Figure VI.165 shows an example of a sequence of iterates $\{T^n f_0 : n = 0, 1, 2, 3, \dots\}$ obtained by repeated application of the contraction mapping T , introduced in the proof of Theorem 2.2. The initial function $f_0(x)$ is linear. The sequence converges to the fractal interpolation function f , which is the fixed point of T . Notice that the whole image can be interpreted as the attractor of an IFS with condensation, where the condensation set is the graph of the function $f_0(x)$.

The reader may wonder, in view of the proof of Theorem 2.2, why we go to the trouble of establishing that there is a metric such that the IFS is contractive. After all, we could simply use T to construct fractal interpolation functions. The answer has two parts, (a) and (b). (a) We can now apply the theory of hyperbolic IFS to fractal interpolation functions. Of special importance, this means that we can use IFS algorithms to compute fractal interpolation functions, that the Collage Theorem can be used as an aid to finding fractal interpolation functions that approximate given data,

and that we can use the Hausdorff metric to discuss the accuracy of approximation of experimental data by a fractal interpolation function. (b) By treating fractal interpolation functions as attractors of IFS of affine transformations we provide a common language for the description of an important class of functions and sets: the same type of formula, namely an IFS code, can be used in all cases.

One consequence of the fact that the IFS $\{\mathbb{R}; w_n, n = 1, 2, \dots, N\}$ associated with a set of data $\{(x_n, F_n) : n = 1, 2, \dots, N\}$ is hyperbolic is that any set $A_0 \in \mathcal{H}(\mathbb{R}^2)$ leads to a Cauchy sequence of sets $\{A_n\}$ that converges to G in the Hausdorff metric. In the usual way we define $W : \mathcal{H}(\mathbb{R}^2) \rightarrow \mathcal{H}(\mathbb{R}^2)$ by

$$W(B) = \cup_{n=1}^N w_n(B) \quad \text{for all } B \in \mathcal{H}(\mathbb{R}^2).$$

Then $\{A_n = W^{on}(A_0)\}$ is a Cauchy sequence of sets which converges to G in the Hausdorff metric. This idea is illustrated in Figure VI.166. Notice that if A_0 is the graph of a function $f_0 \in \mathcal{F}$ then A_n is the graph of $T^{on} f_0$.

Examples & Exercises

2.10. Prove that the metric on $\mathbb{R}^2$ introduced in the proof of Theorem 2.1 is equivalent to the Euclidean metric on $\mathbb{R}^2$.

2.11. Use the Collage Theorem to help you find a fractal interpolation function that approximates the function whose graph is shown in Figure VI.167.

2.12. Write a program that allows you to use the Deterministic Algorithm to compute fractal interpolation functions.

2.13. Explain why Theorems 2.1 and 2.2 have the restriction that N is greater than 1.

2.14. Let a set of data $\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$ be given. Let the metric space $(\mathcal{F}, d)$ and the transformation $T : \mathcal{F} \rightarrow \mathcal{F}$ be defined as in the proof of Theorem 2.2. Prove that if $f \in \mathcal{F}$ then Tf is an interpolation function associated with the data. Deduce that if $f \in \mathcal{F}$ is a fixed point of T then f is an interpolation function associated with the data.

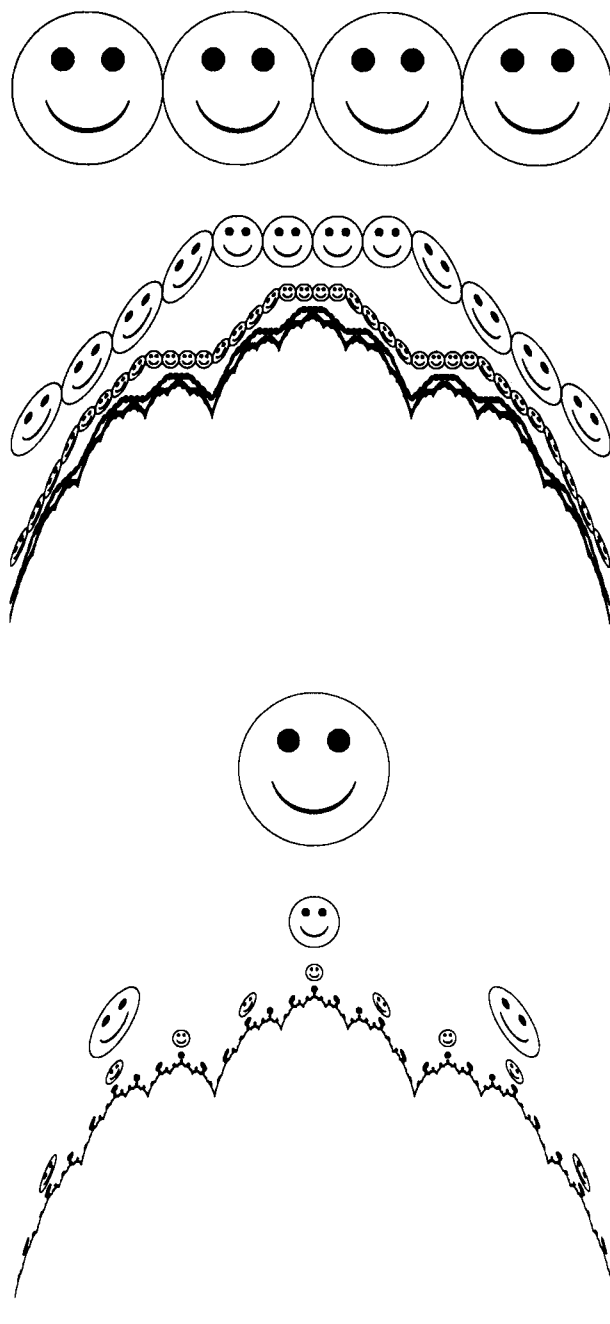
2.15. Make a nonlinear generalization of the theory of fractal interpolation functions. For example, consider what happens if one uses an IFS made up of nonlinear transformations $w_n : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of the form

$$w_n(x, y) = (a_n x + e_n, c_n x + d_n y + g_n y^2 + f_n),$$

where a_n, c_n, d_n, g_n , and f_n are real constants. This example uses “quadratic scaling” in the vertical direction instead of linear scaling. Determine sufficient conditions for the IFS to be hyperbolic, with an attractor that is the graph of a function that interpolates the data $\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$. Note that in certain circumstances the IFS generates the graph of a differentiable interpolation function.

2.16. Let $f(x)$ denote a fractal interpolation function associated with a set of data

Figure VI.166. Examples of the convergence of a sequence of sets $\{A_n\}$ in the Hausdorff metric, to the graph of a fractal interpolation function.



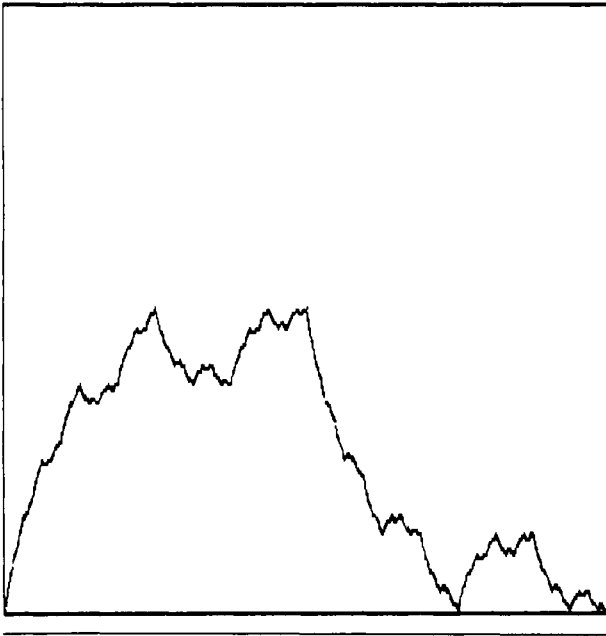


Figure VI.167. Use the Collage Theorem to find an IFS $\{\mathbb{R}^2; w_1, w_2\}$, where w_1 and w_2 are shear transformations on $\mathbb{R}^2$, such that the attractor of the IFS is a good approximation to the graph of the function shown here.

$\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$, where $N > 1$. Let the metric space $(\mathcal{F}, d)$ and the transformation $T : \mathcal{F} \rightarrow \mathcal{F}$ be defined as in the proof of Theorem 2.2. The functional equation $Tf = f$ can be used to evaluate various integrals of f . As an example consider the problem of evaluating the integral

$$I = \int_{x_0}^{x_N} f(x) dx.$$

The integral is well defined because $f(x)$ is continuous. We have

$$\begin{aligned} I &= \int_{x_0}^{x_N} (Tf)(x) dx = \sum_{n=1}^N \int_{x_{n-1}}^{x_n} (Tf)(x) dx \\ &= \sum_{n=1}^N \int_{x_0}^{x_N} (c_n x + d_n f(x) + f_n) d(a_n x + e_n) = \alpha I + \beta, \end{aligned}$$

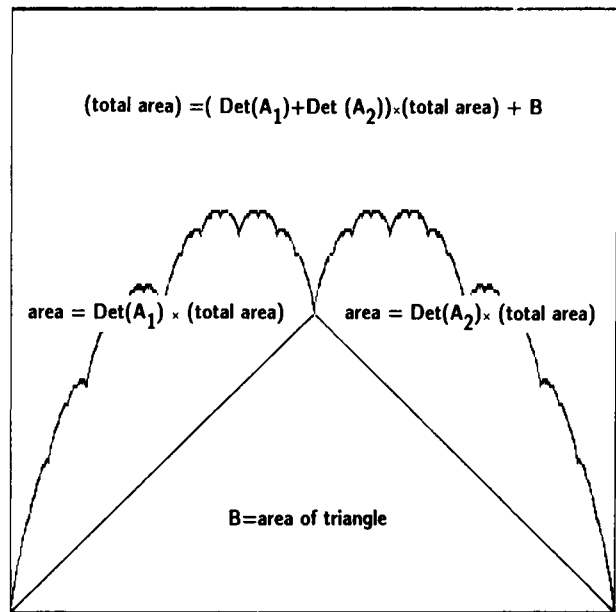
where

$$\alpha = \left(\sum_{n=1}^N a_n d_n \right) \quad \text{and} \quad \beta = \sum_{n=1}^N a_n \int_{x_0}^{x_N} (c_n x + f_n) dx.$$

Show that, under the standard assumptions, $|\alpha| < 1$. Show also that

$$\beta = \int_{x_0}^{x_N} f_0(x) dx,$$

Figure VI.168. Illustration of the geometrical viewpoint concerning the integration of fractal interpolation functions.



where $f_0(x)$ is the piecewise linear interpolation function associated with the data. Conclude that

$$\int_{x_0}^{x_N} f(x) dx = \frac{\beta}{(1 - \alpha)}.$$

Check this result for the case of the parabola, described in Exercise 2.4. In Figure VI.168 we illustrate a geometrical way of thinking about the integration of a fractal interpolation function.

2.17. Let $f(x)$ denote a fractal interpolation function associated with a set of data $\{(x_i, F_i) : i = 0, 1, 2, \dots, N\}$, where $N > 1$. By following similar steps to those in exercise 2.16, find a formula for the integral

$$I_1 = \int_{x_0}^{x_N} x f(x) dx.$$

Check your formula by applying it to the parabola described in exercise 2.4.

2.18. Figure VI.169 shows a fractal interpolation function together with a zoom. Can you reproduce these images and then make a further zoom? What do you expect a very high-magnification zoom to look like?

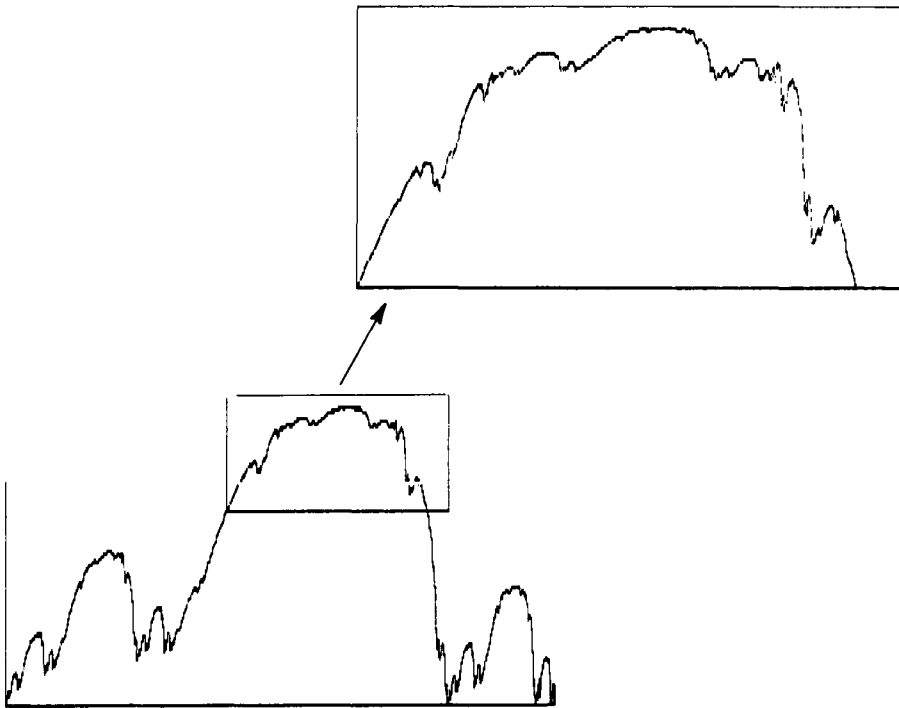


Figure VI.169. A fractal interpolation function together with a zoom. If the fractal dimension is equal to 1, what do you expect “most” very high magnification zooms to look like?

3 The Fractal Dimension of Fractal Interpolation Functions

The following excellent theorem tells us the fractal dimension of fractal interpolation functions.

Theorem 3.1 *Let N be a positive integer greater than 1. Let $\{(x_n, F_n) \in \mathbb{R}^2 : n = 1, 2, \dots, N\}$ be a set of data. Let $\{\mathbb{R}^2; w_n, n = 1, 2, \dots, N\}$ be an IFS associated with the data, where*

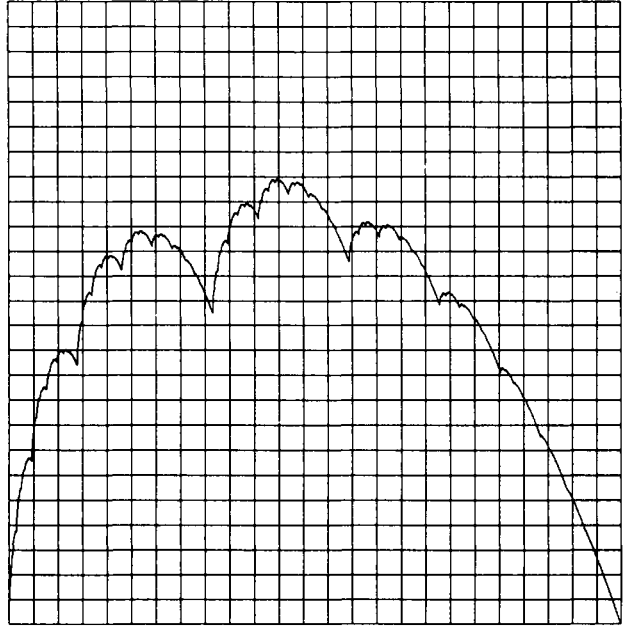
$$w_n \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_n & 0 \\ c_n & d_n \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e_n \\ f_n \end{pmatrix} \quad \text{for } n = 1, 2, \dots, N.$$

The vertical scaling factors d_n obey $0 \leq d_n < 1$; and the constants a_n, c_n, e_n , and f_n are given by equations 1, 2, 3, and 4, for $n = 1, 2, \dots, N$. Let G denote the attractor of the IFS, so that G is the graph of a fractal interpolation function associated with the data. If

$$\sum_{n=1}^N |d_n| > 1 \tag{1}$$

and the interpolation points do not all lie on a single straight line, then the fractal dimension of G is the unique real solution D of

Figure VI.170. The graph G of a fractal interpolation function is superimposed on a grid of closed square boxes of side length ϵ . $\mathcal{N}(\epsilon)$ is used to denote the number of boxes that intersect G . What is the value of $\mathcal{N}(\epsilon)$?



$$\sum_{n=1}^N |d_n| a_n^{D-1} = 1.$$

Otherwise the fractal dimension of G is 1.

Proof (Informal Demonstration). The formal proof of this theorem can be found in [Barnsley 1988f]. Here we give an informal argument for why it is true. We use the notation in the statement of the theorem.

Let $\epsilon > 0$. We consider G to be superimposed on a grid of closed square boxes of side length ϵ , as illustrated in Figure VI.170. Let $\mathcal{N}(\epsilon)$ denote the number of square boxes of side length ϵ which intersect G . These boxes are similar to the ones used in the Box Counting Theorem, Theorem 1.2 in Chapter V, except that their sizes are arbitrary. On the basis of the intuitive idea introduced in Chapter V, section 1, we suppose that G has fractal dimension D , where

$$\mathcal{N}(\epsilon) \approx \text{constant} \cdot \epsilon^{-D} \text{ as } \epsilon \rightarrow 0.$$

We want to estimate the value of D on the basis of this assumption.

Let $n \in \{1, 2, \dots, N\}$. Let $\mathcal{N}_n(\epsilon)$ denote the number of boxes of side length ϵ which intersect $w_n(G)$ for $n = 1, 2, \dots, N$. We suppose that ϵ is very small compared to $|x_N - x_0|$. Then because the IFS is just-touching it is reasonable to make the approximation

$$\mathcal{N}(\epsilon) \approx \mathcal{N}_1(\epsilon) + \mathcal{N}_2(\epsilon) + \mathcal{N}_3(\epsilon) + \dots + \mathcal{N}_N(\epsilon). \quad (2)$$

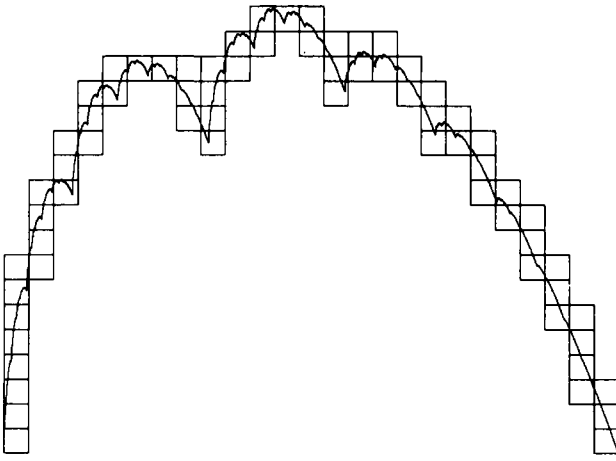


Figure VI.171. The boxes that intersect G can be thought of as organized in columns. The set of columns of boxes of side length ϵ which intersect G is denoted by $\{c_j(\epsilon) : j = 1, 2, \dots, K(\epsilon)\}$, where $K(\epsilon)$ denotes the number of columns. What is the value of $K(\epsilon)$ and how many boxes are there in $c_2(\epsilon)$, in this illustration?

We now look for a relationship between $\mathcal{N}(\epsilon)$ and $\mathcal{N}_n(\epsilon)$. The boxes that intersect G can be thought of as being organized into columns, as illustrated in Figure VI.171.

Let the set of columns of boxes of side length ϵ which intersect G be denoted by $\{c_j(\epsilon) : j = 1, 2, \dots, K(\epsilon)\}$, where $K(\epsilon)$ denotes the number of columns. Under the conditions in equation 1, in the statement of the theorem, one can prove that the minimum number of boxes in a column increases without limit as ϵ approaches zero. To simplify the discussion we assume that

$$|d_n| > a_n \text{ for } n = 1, 2, \dots, N.$$

(Notice that

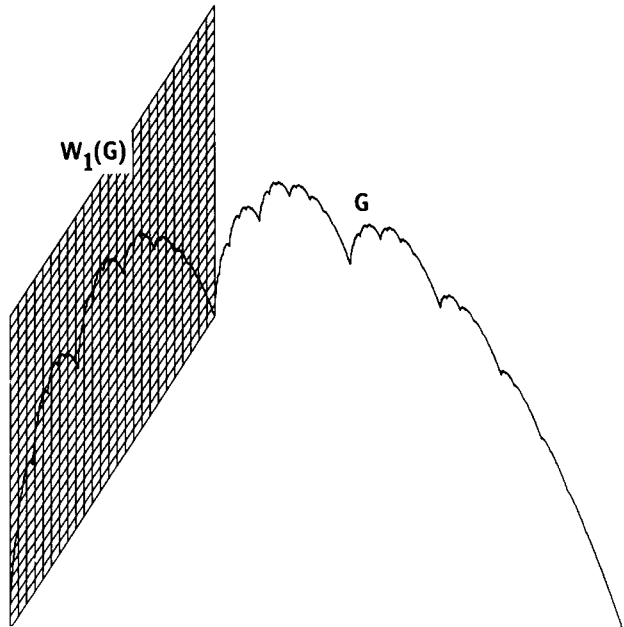
$$\sum_{n=1}^N a_n = \sum_{n=1}^N \frac{(x_n - x_{n-1})}{(x_N - x_0)} = 1,$$

which tells us that this assumption is stronger than the assumption $\sum_{n=1}^N |d_n| > 1$.) Then consider what happens to a column of boxes $c_j(\epsilon)$ of side length ϵ when we apply the affine transformation w_n to it. It becomes a column of parallelograms. The width of the column is $a_n \epsilon$ and the height of the column is $|d_n|$ times the height of the column before transformation. Let $\mathcal{N}(c_j(\epsilon))$ denote the number of boxes in the column $c_j(\epsilon)$. Then the column $w_n(c_j(\epsilon))$ can be thought of as being made up of square boxes of side length $a_n \epsilon$, each of which intersects $w_n(G)$. How many boxes of side length $a_n \epsilon$ are there in this column? Approximately $|d_n| \mathcal{N}(c_j(\epsilon)) / a_n$. Adding up the contribution to $\mathcal{N}_n(a_n \epsilon)$ from each column we obtain

$$\mathcal{N}_n(a_n \epsilon) \approx \sum_{j=1}^{K(\epsilon)} \frac{|d_n| \mathcal{N}(c_j(\epsilon))}{a_n} = \frac{|d_n|}{a_n} \sum_{j=1}^{K(\epsilon)} \mathcal{N}(c_j(\epsilon)) = \frac{|d_n|}{a_n} \mathcal{N}(\epsilon).$$

The situation is illustrated in Figure VI.172.

Figure VI.172. When the shear transformation w_1 is applied to the columns of boxes which cover the graph, G , the result is a set of thinner columns, of width $a_1\epsilon$, which cover $w_1(G)$. The new columns are made up of small parallelograms, but the number of square boxes of side length $a_1\epsilon$ which they contain is readily estimated.



From the last equation we deduce that when ϵ is very small compared to $[x_0, x_N]$,

$$\mathcal{N}_n(\epsilon) \approx \frac{|d_n|}{a_n} \mathcal{N}\left(\frac{\epsilon}{a_n}\right) \text{ for } n = 1, 2, \dots, N. \quad (3)$$

We now substitute from equation 3 into 2 to obtain the functional equation

$$\mathcal{N}(\epsilon) \approx \frac{d_1}{a_1} \mathcal{N}\left(\frac{\epsilon}{a_1}\right) + \frac{d_2}{a_2} \mathcal{N}\left(\frac{\epsilon}{a_2}\right) + \frac{d_3}{a_3} \mathcal{N}\left(\frac{\epsilon}{a_3}\right) + \dots + \frac{d_N}{a_N} \mathcal{N}\left(\frac{\epsilon}{a_N}\right).$$

Into this equation we substitute our assumption $\mathcal{N}(\epsilon) \approx \text{constant} \cdot \epsilon^{-D}$ to obtain the equation

$$\epsilon^{-D} \approx |d_1|a_1^{D-1}\epsilon^{-D} + |d_2|a_2^{D-1}\epsilon^{-D} + |d_3|a_3^{D-1}\epsilon^{-D} + \dots + |d_N|a_N^{D-1}\epsilon^{-D}.$$

The main formula in the statement of the theorem follows at once.

If the interpolation points are collinear, then the attractor of the IFS is the line segment that connects the point (x_0, F_0) to the point (x_N, F_N) , and this has fractal dimension 1. If $\sum_{n=1}^N |d_n| \leq 1$, then one can show that $\mathcal{N}(\epsilon)$ behaves like a constant times ϵ^{-1} , whence the fractal dimension is 1. This completes our informal demonstration of the theorem.

Examples & Exercises

3.1. We consider the fractal dimension of a fractal interpolation function in the case where the interpolation points are equally spaced. Let $x_i = x_0 + \frac{i}{N}(x_N - x_0)$ for $i = 0, 1, 2, \dots, N$. Then it follows that $a_n = \frac{1}{N}$ for $n = 1, 2, \dots, N$. Hence if condition (1) in Theorem 3.1 holds then the fractal dimension D of the interpolation

function obeys

$$\sum_{n=1}^N |d_n| \left(\frac{1}{N}\right)^{D-1} = \left(\frac{1}{N}\right)^{D-1} \sum_{n=1}^N |d_n| = 1.$$

It follows that

$$D = 1 + \frac{\log \left(\sum_{n=1}^N |d_n| \right)}{\log(N)}.$$

This is a delightful formula for reasons of two types, (a) and (b). (a) This formula confirms our understanding of the fractal dimension of fractal interpolation functions. For example, notice that $\sum_{n=1}^N |d_n| < N$. Hence the dimension of a fractal interpolation function is less than 2: however, we can make it arbitrarily close to 2. Also, under the assumption that $\sum_{n=1}^N |d_n| > 1$, the fractal dimension is greater than 1: however, we can vary it smoothly down to 1. (b) It is remarkable that the fractal dimension does not depend on the values $\{F_i : i = 0, 1, 2, \dots, N\}$, aside from the constraint that the interpolation points be noncollinear. Hence it is easy to explore a collection of fractal interpolation functions, all of which have the same fractal dimension, by imposing the following simple constraint on the vertical scaling factors:

$$\sum_{n=1}^N |d_n| = N^{D-1}.$$

Figure VI.173 illustrates some members of the family of fractal interpolation functions corresponding to the set of data $\{(0, 0), (1, 1), (2, 1), (3, 2)\}$, such that the fractal dimension of each member of the family is $D = 1.3$.

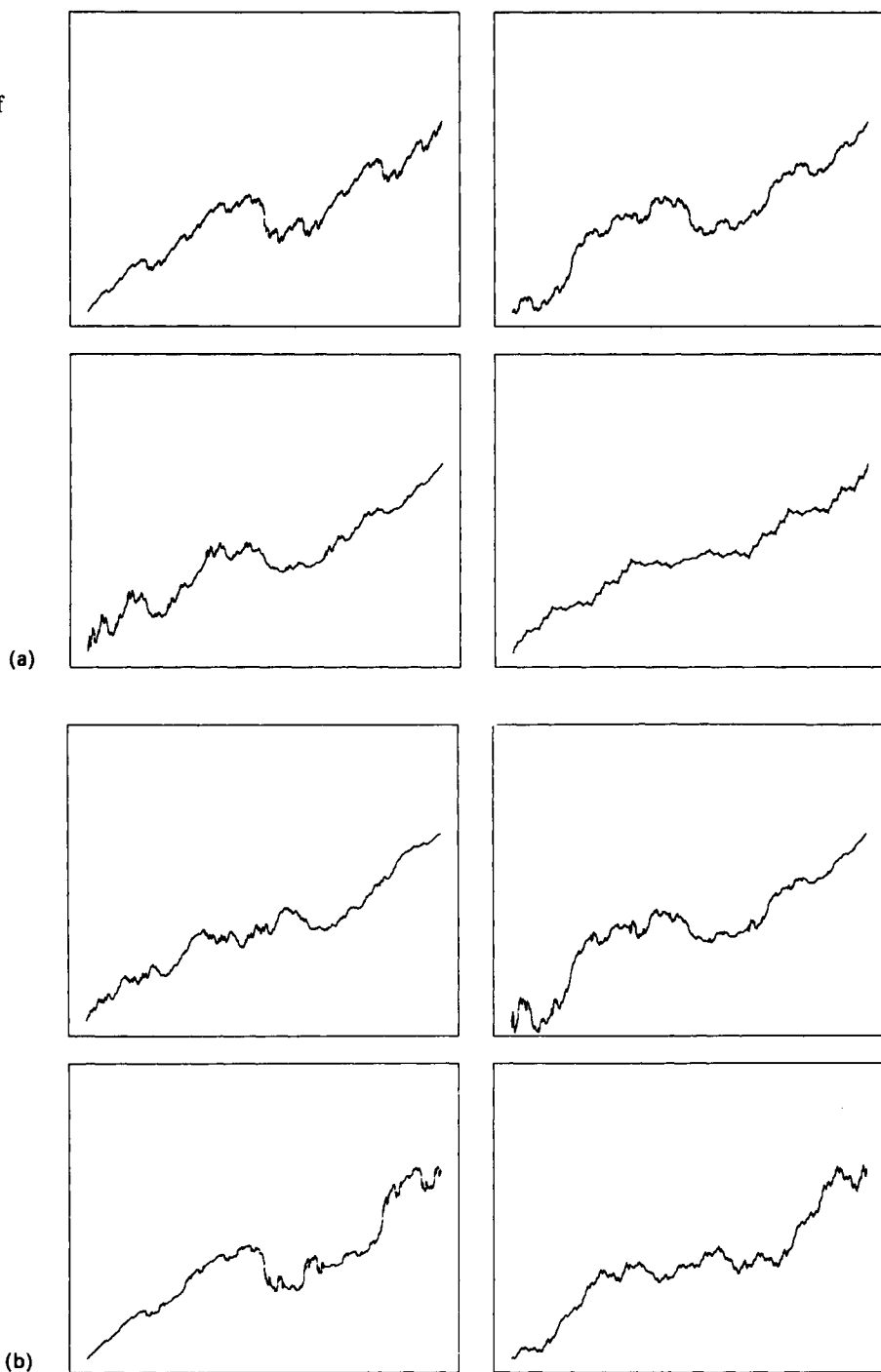
Figures 173 (a) and (c) illustrate members of a family of fractal interpolation functions parameterized by the fractal dimension D . Each function interpolates the same set of data.

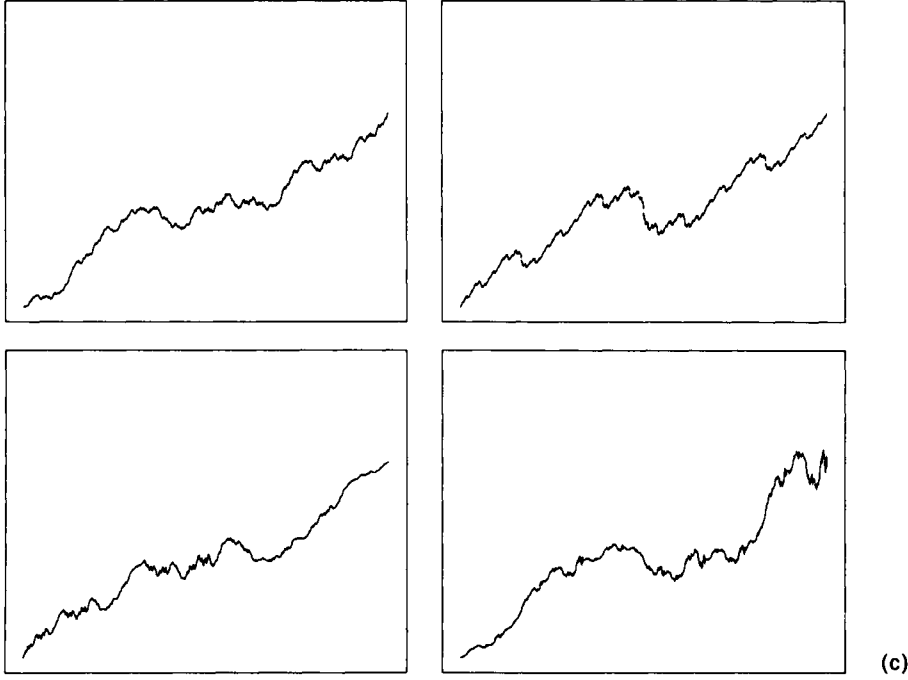
3.2. Make an experimental estimate of the fractal dimension of the graphical data in Figure VI.175. Find a fractal interpolation function associated with the data $\{(0, 0), (50, 50), (100, 0)\}$, which has the same fractal dimension and two equal vertical scaling factors. Compare the graph of the fractal interpolation function with the graphical data.

3.3. Find a fractal interpolation function that approximates the experimental data shown in Figure V.147.

3.4. Figure VI.176 shows the graphs of functions belonging to various one-parameter families of fractal interpolation functions. Each graph is the attractor of an IFS consisting of two affine transformations. Find the IFS associated with one of the families.

Figure VI.173. Members of the family of fractal interpolation functions corresponding to the set of data $\{(0, 0), (1, 1), (2, 1), (3, 2)\}$, such that the fractal dimension of each member of the family is $D = 1.3$.





4 Hidden Variable Fractal Interpolation

We begin by generalizing the results of section 6.2. Throughout this section, let (Y, d_Y) denote a complete metric space.

Definition 4.1 Let $I \subset \mathbb{R}$. Let $f : I \rightarrow Y$ be a function. The graph of f is the set of points

$$G = \{(x, f(x)) \in \mathbb{R} \times Y : x \in I\}.$$

Definition 4.2 A set of generalized data is a set of points of the form $\{(x_i, F_i) \in \mathbb{R} \times Y : i = 0, 1, 2, \dots, N\}$, where

$$x_0 < x_1 < x_2 < x_3 < \dots < x_N.$$

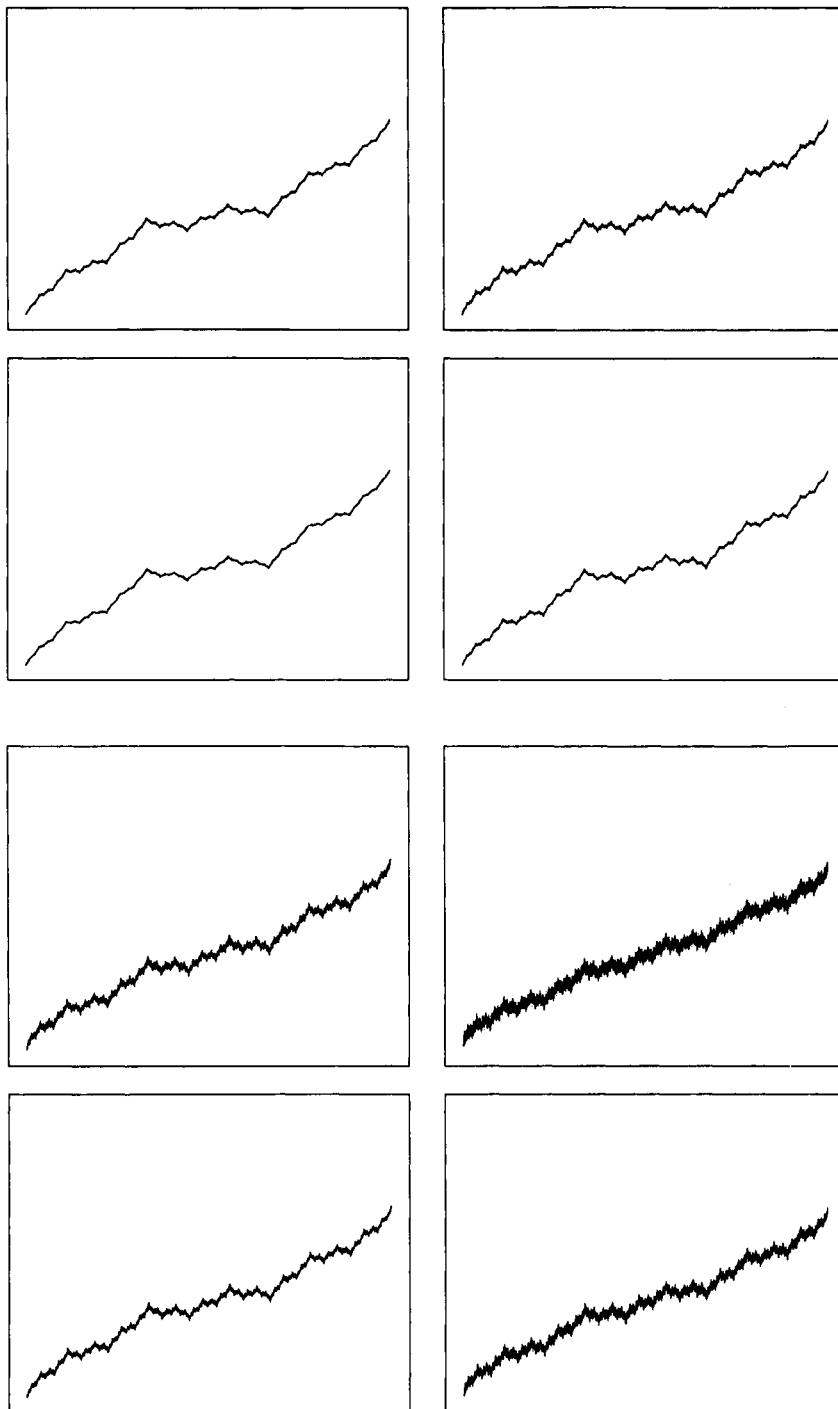
An interpolation function corresponding to this set of data is a continuous function $f : [x_0, x_N] \rightarrow Y$ such that

$$f(x_i) = F_i \quad \text{for } i = 1, 2, \dots, N.$$

The points $(x_i, F_i) \in \mathbb{R} \times Y$ are called the interpolation points. We say that the function f interpolates the data and that (the graph of) f passes through the interpolation points.

Let X denote the Cartesian product space $\mathbb{R} \times Y$. Let θ denote a positive number.

Figure VI.174. Members of a one-parameter family of fractal interpolation functions. They correspond to the set of data $\{(0, 0), (1, 1), (2, 1), (3, 2)\}$ with vertical scaling factors $d_1 = -d_2 = d_3 = 3^{D-2}$ for $D = 1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6$, and 1.7 . D is the fractal dimension of the fractal interpolation function.



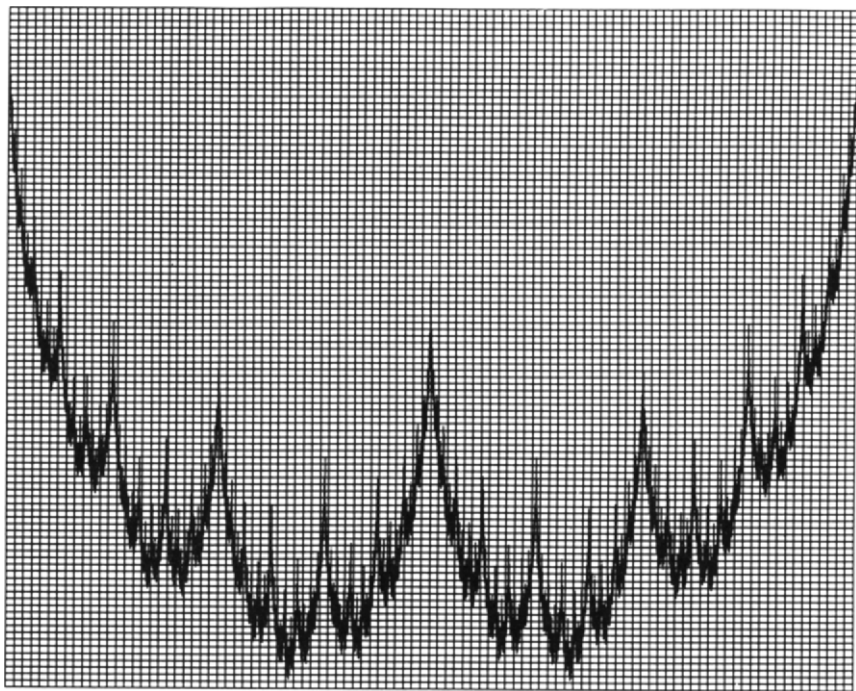


Figure VI.175. Make an experimental estimate of the fractal dimension of the graphical data shown here. Find a fractal interpolation function associated with the data $\{(0, 0), (50, -50), (100, 0)\}$, which has the same fractal dimension and two equal vertical scaling factors. Compare the graph of the fractal interpolation function with the graphical data.

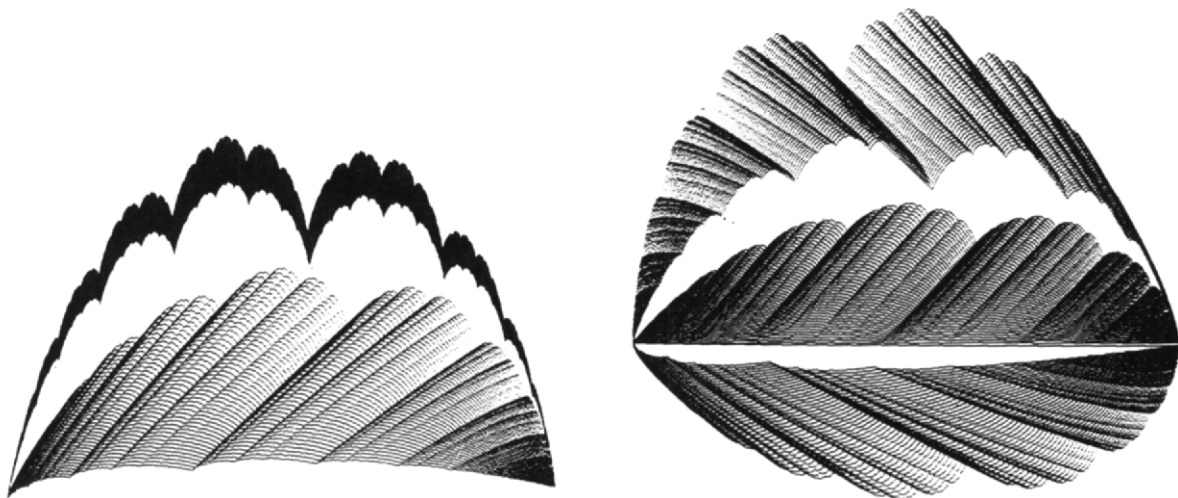


Figure VI.176. This figure shows the graphs of various one-parameter families of fractal interpolation functions. Each graph is the attractor of an IFS consisting of two affine transformations. Can you find the families?

Define a metric d on $\mathbf{X}$ by

$$d(X_1, X_2) = |x_1 - x_2| + \theta d_Y(y_1, y_2), \quad (1)$$

for all points $X_1 = (x_1, y_1)$ and $X_2 = (x_2, y_2)$ in $\mathbf{X}$. then $(\mathbf{X}, d)$ is a complete metric space.

Let N be an integer greater than 1. Let a set of generalized data $\{(x_i, F_i) \in \mathbf{X} : i = 0, 1, 2, \dots, N\}$ be given. Let $n \in \{1, 2, \dots, N\}$. Define $L_n : \mathbb{R} \rightarrow \mathbb{R}$ by

$$L_n(x) = a_n x + e_n \text{ where } a_n = \frac{(x_n - x_{n-1})}{(x_N - x_0)} \text{ and } e_n = \frac{(x_N x_{n-1} - x_0 x_n)}{(x_N - x_0)}. \quad (2)$$

so that $L_n([x_0, x_N]) = [x_{n-1}, x_n]$. Let c and s be real numbers, with $0 \leq s < 1$ and $c > 0$. For each $n \in \{1, 2, \dots, N\}$ let $M_n : \mathbf{X} \rightarrow \mathbf{Y}$ be a function that obeys

$$d(M_n(a, y), M_n(b, y)) \leq c|a - b| \text{ for all } a, b \in \mathbb{R}, \quad (3)$$

and

$$d(M_n(x, a), M_n(x, b)) \leq s d_Y(a, b) \text{ for all } a, b \in \mathbf{Y}. \quad (4)$$

Define a transformation $w_n : \mathbf{X} \rightarrow \mathbf{X}$ by

$$w_n(x, y) = (L_n(x), M_n(x, y)) \text{ for all } (x, y) \in \mathbf{X}, n = 1, 2, \dots, N.$$

Theorem 4.1 *Let the IFS $\{\mathbf{X}; w_n, n = 1, 2, \dots, N\}$ be defined as above with $N > 1$. In particular, assume that there are real constants c and s such that $0 \leq s \leq 1$, $0 < c$, and conditions 3 and 4 are obeyed. Let the constant θ in the definition of the metric d in equation 1 be defined by*

$$\theta = \frac{(1 - a)}{2c} \text{ where } a = \max\{a_i : i = 1, 2, \dots, N\}.$$

Then the IFS $\{\mathbf{X}; w_n, n = 1, 2, \dots, N\}$ is hyperbolic with respect to the metric d .

Proof This follows very similar lines to the proof of Theorem 2.1. We leave it as an exercise for enthusiastic readers. The proof can also be found in [Barnsley 1986].

We now constrain the hyperbolic IFS $\{\mathbf{X}; w_n, n = 1, 2, \dots, N\}$, defined above, to ensure that its attractor includes the set of generalized data. We assume that

$$M_n(x_0, F_0) = F_{n-1} \text{ and } M_n(x_N, F_N) = F_n \text{ for } n = 1, 2, \dots, N. \quad (5)$$

Then it follows that

$$w_n(x_0, F_0) = (x_{n-1}, F_{n-1}) \text{ and } w_n(x_N, F_N) = (x_n, F_n) \text{ for } n = 1, 2, \dots, N.$$

Theorem 4.2 *Let N be a positive integer greater than 1. Let $\{\mathbf{X}; w_n, n = 1, 2, \dots, N\}$ denote the IFS defined above, associated with the generalized data set $\{(x_i, F_i) \in \mathbb{R} \times \mathbf{Y} : i = 1, 2, \dots, N\}$. In particular, assume that there are real constants c and s such that $0 \leq s \leq 1$, $0 < c$, and conditions 3, 4, and 5 are obeyed. Let*

$G \in \mathcal{H}(X)$ denote the attractor of the IFS. Then G is the graph of a continuous function $f : [x_0, x_N] \rightarrow Y$ which interpolates the data $\{(x_i, F_i) : i = 1, 2, \dots, N\}$. That is,

$$G = \{(x, f(x)) : x \in [x_0, x_N]\},$$

where

$$f(x_i) = F_i \quad \text{for } i = 0, 1, 2, 3, \dots, N.$$

Proof Again we refer to [Barnsley 1988f]. The proof is analogous to the proof of Theorem 2.2.

Definition 4.3 The function whose graph is the attractor of an IFS, as described in Theorems 4.1 and 4.2, above, is called a generalized fractal interpolation function, corresponding to the generalized data $\{(x_i, F_i) : i = 1, 2, \dots, N\}$.

We now show how to use the idea of generalized fractal interpolation functions to produce interpolation functions that are more flexible than heretofore. The idea is to construct a generalized fractal interpolation function, using affine transformations acting on $\mathbb{R}^3$, and to project its graph into $\mathbb{R}^2$. This can be done in such a way that the projection is the graph of a function that interpolates a set of data $\{(x_i, F_i) \in \mathbb{R}^2 : i = 1, 2, \dots, N\}$. The extra degrees of freedom provided by working in $\mathbb{R}^3$ give us “hidden” variables. These variables can be used to adjust the shape and fractal dimension of the interpolation functions. The benefits of working with affine transformations are kept.

Let N be an integer greater than 1. Let a set of data $\{(x_i, F_i) \in \mathbb{R}^2 : i = 0, 1, 2, \dots, N\}$ be given. Introduce a set of real parameters $\{H_i : i = 0, 1, 2, \dots, N\}$. For the moment let us suppose that these parameters are fixed. Then we define a generalized set of data to be $\{(x_i, F_i, H_i) \in \mathbb{R} \times \mathbb{R}^2 : i = 0, 1, 2, \dots, N\}$. In the present application of Theorem 4.2 we take (Y, d_Y) to be $(\mathbb{R}^2, \text{Euclidean})$. We consider an IFS $\{\mathbb{R}^3; w_n, n = 1, 2, \dots, N\}$, where for $n \in \{1, 2, \dots, N\}$ the map $w_n : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an affine transformation of the special structure:

$$w_n \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a_n & 0 & 0 \\ c_n & d_n & h_n \\ k_n & l_n & m_n \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} e_n \\ f_n \\ g_n \end{bmatrix}.$$

Here $a_n, c_n, d_n, e_n, f_n, g_n, h_n, k_n, l_n$, and m_n are real numbers. We assume that they obey the constraints

$$w_n \begin{bmatrix} x_0 \\ F_0 \\ G_0 \end{bmatrix} = w_n \begin{bmatrix} x_{n-1} \\ F_{n-1} \\ G_{n-1} \end{bmatrix},$$

and

$$w_n \begin{bmatrix} x_N \\ F_N \\ G_N \end{bmatrix} = w_n \begin{bmatrix} x_n \\ F_n \\ G_n \end{bmatrix}, \text{ for } n = 1, 2, \dots, N.$$

Then we can write

$$w_n(x, y, z) = (L_n(x), M_n(x, y, z)) \text{ for all } (x, y, z) \in \mathbb{R}^3, n = 1, 2, \dots, N,$$

where $L_n(x)$ is defined in Equation 2 and $M_n : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by

$$M_n \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A_n \begin{bmatrix} y \\ z \end{bmatrix} + \begin{bmatrix} f_n + c_n x \\ g_n + k_n x \end{bmatrix},$$

where

$$A_n = \begin{bmatrix} d_n & h_n \\ l_n & m_n \end{bmatrix} \text{ for } n = 1, 2, \dots, N. \quad (6)$$

Let us replace F_n in condition 5 by (F_n, H_n) . Then M_n obeys condition 5. Let us define

$$c = \max\{\max\{c_i, k_i\} : i = 1, 2, \dots, N\}.$$

Then condition 3 is true. Lastly, assume that the linear transformation $A_n : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is contractive with contractivity factor s with $0 \leq s < 1$. Then condition 4 is true. We conclude that, under the conditions given in this paragraph, the IFS $\{\mathbb{R}^3; w_n, n = 1, 2, \dots, N\}$ satisfies the conditions of Theorem 4.2. It follows that the attractor of the IFS is the graph of a continuous function $f : [x_0, x_N] \rightarrow \mathbb{R}^2$ such that

$$f(x_i) = (F_i, H_i) \text{ for } i = 1, 2, \dots, N.$$

Now write

$$f(x) = (f_1(x), f_2(x)).$$

Then $f_1 : [x_0, x_N] \rightarrow \mathbb{R}$ is a continuous function such that

$$f_1(x_i) = F_i \text{ for } i = 1, 2, \dots, N.$$

Definition 4.4 The function $f_1 : [x_0, x_N] \rightarrow \mathbb{R}^2$ constructed in the previous paragraph is called a hidden variable fractal interpolation function, associated with the set of data $\{(x_i, F_i) \in \mathbb{R}^2 : i = 1, 2, \dots, N\}$.

The easiest method for computing the graph of a hidden variable fractal interpolation function is with the aid of the Random Iteration Algorithm. Here we present an adaptation of Program 1. It computes points on the graph of a hidden variable fractal interpolation function and displays them on a graphics monitor. It is written for $N = 3$ and the data set

$$\{(0, 0), (30, 50), (60, 40), (100, 10)\}.$$

The "hidden" variables, namely the entries of the matrices A_n and the number H_n for $n = 1, 2, 3$, are input by the user during execution of the code. The program calculates the coefficients in the three-dimensional affine transformations from the data, and then applies the Random Iteration Algorithm to the resulting IFS. The first two coordinates of each successively computed point, which has three coordinates, are plotted on the screen of the graphics monitor. The program is written in BASIC. It runs without modification on an IBM PC with Enhanced Graphics Adaptor and Turbobasic. On any line the words preceded by a ' are comments: they are not part of the program.

Program 2.

```
x[0] =0 : x[1] =30 : x[2] =60 : x[3] =100 'Data set

F[0] =0 : F[1] =50 : F[2] =40 : F[3] =10

input "enter the hidden variables H[0], H[1], H[2] and H[3]",
      H[1],H[2],H[3],H[4] 'Hidden Variables

for n = 1 to 3 : print "for n = ", n

input "enter the hidden variables d, h, l, m",
      d[n],hh[n],l[n],m[n] 'More Hidden Variables

next

for n = 1 to 3 'Calculate the affine transformations from
      the Data and the Hidden Variables

p = F[n-1]-d[n]*F[0]-hh[n]*H[0] : q = H[n-1]-l[n]*F[0]-m[n]*H[0]

r = F[n]-d[n]*F[3]-hh[n]*H[3] : s = H[n]-l[n]*F[3]-m[n]*H[3]

      b = x[3]-x[0] : c[n] = (r-p)/b : k[n] = (s-q)/b

a[n] = (x[n]-x[n-1])/b : e[n] = (x[3]*x[n-1]-x[0]*x[n])/b

ff[n] = p-c[n]*x[0] : g[n] = q-k[n]*x[0]

next

screen 2 : cls 'inititalize graphics

window(0,0)-(100,100) 'change this to zoom and/or pan
```

```

x =0: y =0: z =hh[0]  'initial point from which the random
                      iteration begins

for n = 1 to 1000 'Random Iteration Algorithm

    kk = int(3*rnd-0.0001)+1

    newx =  a[kk]*x + e[kk]

    newy =  c[kk]*x + d[kk]*y + hh[kk]*z + ff[kk]

    newz =  k[kk]*x + l[kk]*y + m[kk]*z + g[kk]

    x = newx : y = newy: z = newz

    pset(x,y),z  'plot the most recently computed point, in
                  color z, on the screen

next

end

```

The result of running an adaptation of this program on a Masscomp workstation and then printing the contents of the graphics screen is presented in Figure VI.177. In this case $H[0] = 0$, $H[1] = 30$, $H[2] = 60$, $H[3] = 100$, $d(1) = d(2) = d(3) = 0.3$, $h(1) = h(2) = 0.2$, $h(3) = 0.1$, $l(1) = l(2) = l(3) = -0.1$, $m(1) = 0.3$, $m(2) = 0$, $m(3) = -0.1$. Remember that the linear transformation A_n must be contractive, so certainly do not enter values of magnitude larger than 1 for any of the numbers $d(n)$, $h(n)$, $l(n)$, and $m(n)$. The program renders each point in a color that depends on its z -coordinate. This helps the user to visualize the “hidden” three-dimensional character of the curve.

The important point about hidden variable fractal interpolation is this. Although the attractor of the IFS is a union of affine transformations applied to the attractor, this is not the case in general when we replace the word “attractor” by the phrase “projection of the attractor.” The graph of the hidden variable fractal interpolation function $f_1(x)$ is not self-similar, or self-affine, or self-anything!

The idea of hidden variable fractal interpolation functions can be developed using any number of “hidden” dimensions. As the number of dimensions is increased, the process of specifying the function becomes more and more onerous, and the function itself, seen by us in flatland, becomes more and more random. One would never guess, from looking at pictures of them, that they are generated by *deterministic fractal geometry*.

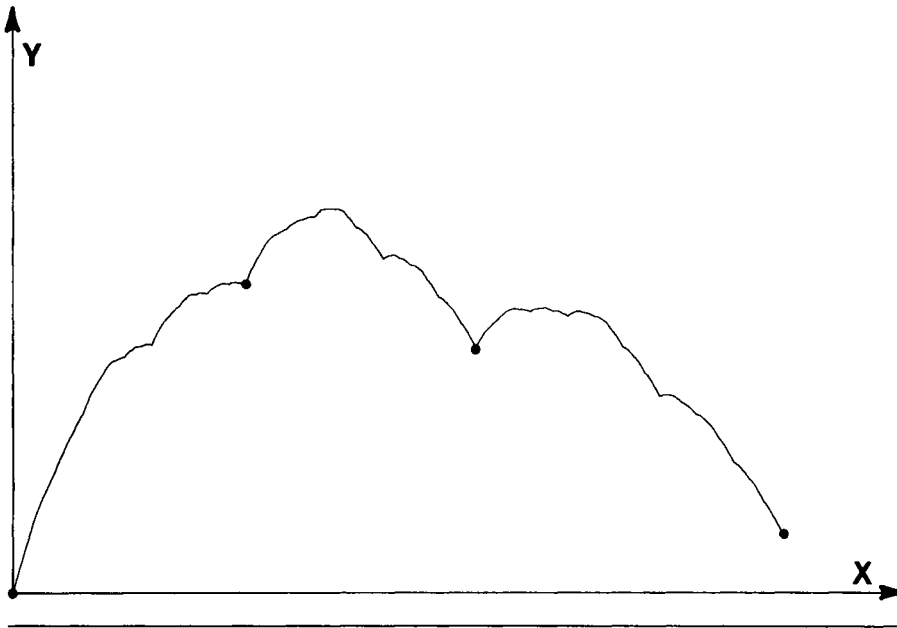


Figure VI.177. An example of a hidden variable fractal interpolation function. This graph was computed using Program 2 with the following “hidden” variables: $H[0] = 0$, $H[1] = 30$, $H[2] = 60$, $H[3] = 100$, $d(1) = d(2) = d(3) = 0.3$, $h(1) = h(2) = 0.2$, $h(3) = 0.1$, $l(1) = l(2) = l(3) = -0.1$, $m(1) = 0.3$, $m(2) = 0$, $m(3) = -0.1$.

Examples & Exercises

4.1. Generalize the proof of Theorem 2.1 to obtain a proof of Theorem 4.1.

4.2. Let $\mathcal{F}$ denote the set of continuous functions $f : [x_0, x_N] \rightarrow \mathbf{Y}$ such that

$$f(x_0) = F_0 \text{ and } f(x_N) = F_N.$$

Define a metric d on $\mathcal{F}$ by

$$d(f, g) = \max\{d_Y(f(x), g(x)) : x \in [x_0, x_N]\}.$$

Then $(\mathcal{F}, d)$ is a complete metric space; see, for example [Rudin 1966]. Use this fact to help you generalize the proof of Theorem 2.2 to provide a proof of Theorem 4.2.

4.3. Rewrite Program 2 in a form suitable for your own computer environment, then run it and obtain hardcopy of the output.

4.4. Modify your version of Program 2 so that you can adjust one of the “hidden” variables while it is running. In this way, make a picture that shows a one-parameter family of hidden variable fractal interpolation functions.

4.5. Modify your version of Program 2 so that you can see the projection of the attractor of the IFS into the (y, z) plane. To do this simply plot (y, z) in place of (x, y) . Make hardcopy of the output.

4.6. Figure VI.178 shows three projections of the graph G of a generalized fractal interpolation function $f : [0, 1] \rightarrow \mathbb{R}^2$. The projections are (i) into the (x, y) plane, (ii) into the (x, z) plane, and (iii) into the (y, z) plane. G is the attractor of an IFS of

the form $\{\mathbb{R}^3; w_1, w_2\}$, where w_1 and w_2 are affine transformations. Find w_1 and w_2 . See also Figure VI.179.

4.7. Use a hidden-variable fractal interpolation function to fit the experimental data in Figure V.147. Here is one way to proceed. (a) Modify your version of Program 6.4.1 so that you can adjust the “hidden” variables from the keyboard. (b) Trace the data in the Figure V.147 onto a sheet of flexible transparent material, such as a viewgraph. (c) Attach the tracing to the screen of your graphics monitor using clear sticky tape. (d) Interactively adjust the “hidden” variables to provide a good visual fit to the data.

4.8. ★Show that, with hidden variables, one can use affine transformations to construct graphs of polynomials of any degree.

5 Space-Filling Curves

Here we make a delightful application of Theorem 4.2. Let A denote a nonempty pathwise-connected compact subset of $\mathbb{R}^2$. We show how to construct a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1]) = A$.

Let (Y, d_Y) denote the metric space $(\mathbb{R}^2, \text{Euclidean})$. We represent points in Y using a Cartesian coordinate system defined by a y -axis and a z -axis. Thus, (y, z) may represent a point in Y . To motivate the development we take $A = \blacksquare \subset Y$. Consider the just-touching IFS $\{Y; w_1, w_2, w_3, w_4\}$, where the maps are similitudes of scaling factor 0.5, corresponding to the collage in Figure VI.180.

Let

$$\begin{aligned} (F_0, H_0) &= (0, 0), & (F_1, H_1) &= (0, 0.5), & (F_2, H_2) &= (0.5, 0.5), \\ (F_3, H_3) &= (1, 0.5), & \text{and} & & & \\ (F_4, H_4) &= (1, 0). \end{aligned}$$

The maps are chosen so that

$$w_n(F_0, H_0) = (F_{n-1}, H_{n-1}) \quad \text{and} \quad w_n(F_4, H_4) = (F_n, H_n) \quad \text{for } n = 1, 2, 3, 4.$$

The IFS code for this IFS is given in Table VI.1.

Let $A_0 \in \mathcal{H}(\blacksquare)$ denote a simple curve that connects the point (F_0, H_0) to the point (F_4, H_4) , such that $A_0 \cap \partial\blacksquare = \{(F_0, H_0), (F_4, H_4)\}$. This last condition says that the curve lies in the interior of the unit square box, except for the two endpoints of the curve. Consider the sequence of sets $\{A_n = W^{on}(A_0)\}_{n=0}^\infty$ where $W : \mathcal{H}(\blacksquare) \rightarrow \mathcal{H}(\blacksquare)$ is defined by

$$W(B) = \cup_{n=1}^4 w_n(B) \quad \text{for all } B \in \mathcal{H}(\blacksquare).$$

It follows from Theorem 7.1 in Chapter III that the sequence converges to $\blacksquare$ in the Hausdorff metric. The reader should verify that, for each $n = 1, 2, \dots, A_n$ is a

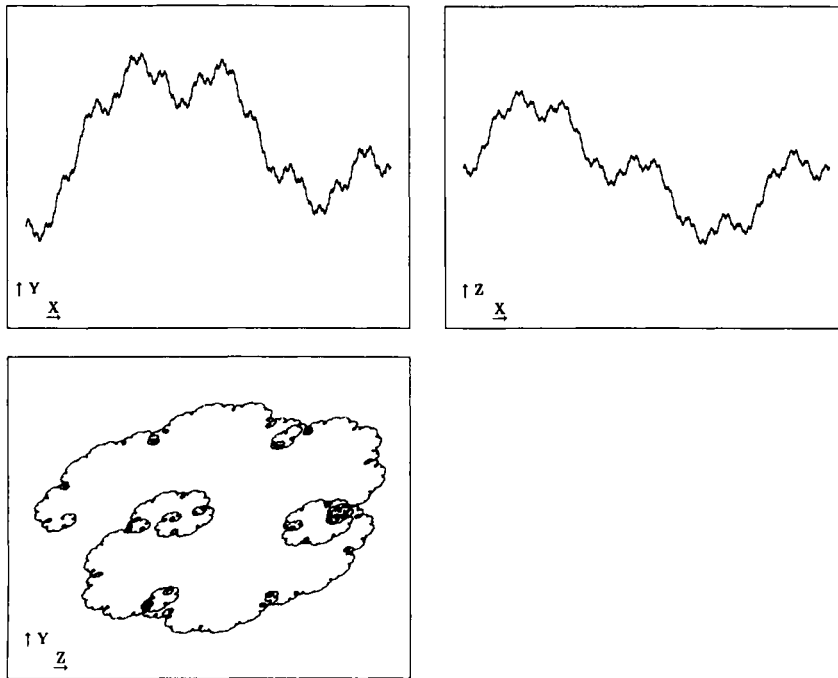


Figure VI.178. This figure shows three projections of the graph of a generalized fractal interpolation function $f : [0, 1] \rightarrow \mathbb{R}^2$. The projections are into the (x, y) plane, the (x, z) plane, and the (y, z) plane. G is the attractor of an IFS of the form $\{\mathbb{R}^3; w_1, w_2\}$, where w_1 and w_2 are affine transformations. Can you find w_1 and w_2 ?

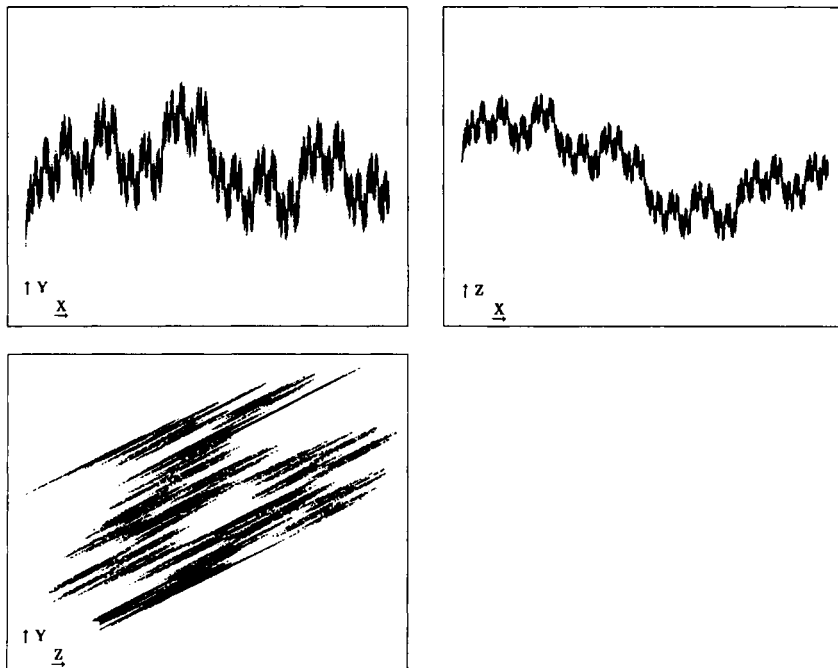


Figure VI.179. Three orthogonal projections of the graph of a generalized fractal interpolation function. The fractal dimension here is higher than for Figure VI.178.

Figure VI.180. Collage of $\blacksquare$ using four similitudes of scaling factor 0.5. The map w_n is chosen so that $w_n(F_0, H_0) = (F_{n-1}, H_{n-1})$ and $w_n(F_4, H_4) = (F_n, H_n)$ for $n = 1, 2, 3, 4$.

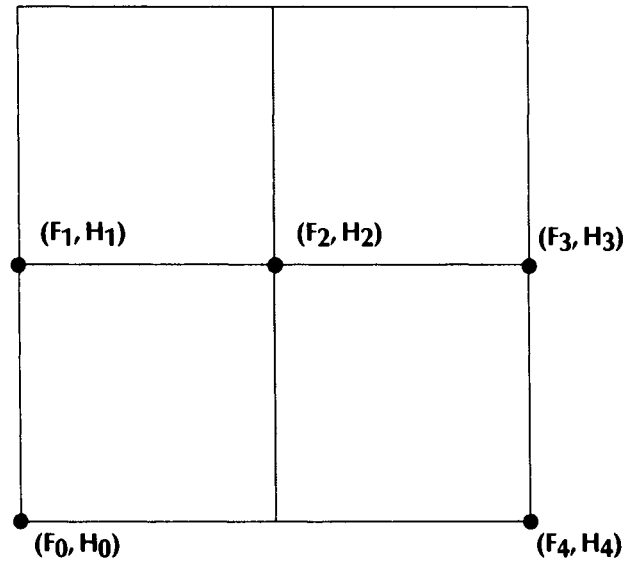


Table VI.1. IFS code for $\blacksquare$, constrained to yield a space-filling curve.

w	a	b	c	d	e	f	p
1	0	0.5	0.5	0	0	0	0.25
2	0.5	0	0	0.5	0	0.5	0.25
3	0.5	0	0	0.5	0.5	0.5	0.25
4	0	-0.5	-0.5	0	1	0.5	0.25

simple curve that connects the points (F_0, H_0) to the point (F_4, H_4) . Sequences of such curves are illustrated in Figures VI.181–VI.184.

We use the IFS defined in the previous paragraph to construct a continuous function $f : [0, 1] \rightarrow \blacksquare$ such that $f([0, 1]) = \blacksquare$. We achieve this by exploiting a hidden variable fractal function constructed in a special way. We use ideas presented in Chapter VI, section 4. Consider the IFS $\{\mathbb{R}^3; w_n, n = 1, 2, \dots, N\}$, where the map $w_n : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the affine transformation

$$w_n \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0.25 & 0 & 0 \\ 0 & a_n & b_n \\ 0 & c_n & d_n \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \begin{bmatrix} (n-1)/4 \\ e_n \\ f_n \end{bmatrix} \text{ for } n \in \{1, 2, 3, 4\}.$$

The constants a_n, b_n, c_n, d_n, e_n , and f_n are defined in Table VI.1. This IFS satisfies Theorem 4.2, corresponding to the set of data

$$\{(0, F_0, H_0), (0.25, F_1, H_1), (0.5, F_2, H_2), (0.75, F_3, H_3), (1, F_4, H_4)\}.$$

It follows that the attractor of the IFS is the graph, G , of a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$. What is the range of this function? It is

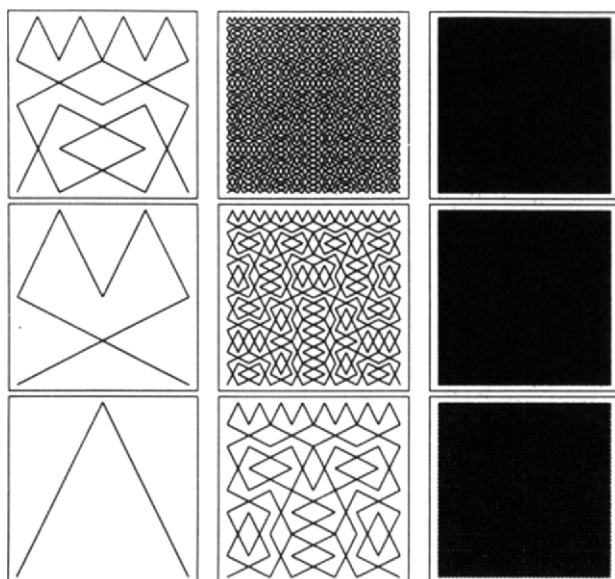


Figure VI.181. A sequence of curves “converging to” a space-filling curve. These are obtained by application of the Deterministic Algorithm to the IFS code in Table VI.1, starting from a curve A_0 , which connects $(0, 0)$ to $(1, 0)$ and lies in ■.

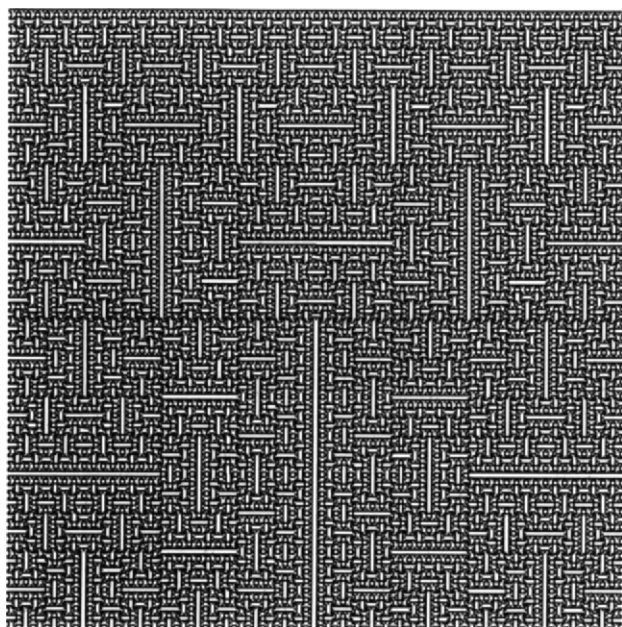


Figure VI.182. A higher-resolution view of one of the panels in Figure VI.184. How long is the shortest path from the lower left corner to the lower right corner?

Figure VI.183. A sequence of sets “converging to” a ■. These are obtained by application of the Deterministic Algorithm to the IFS code in Table VI.1, starting from the set A_0 in the lower left panel. How fascinating they are!

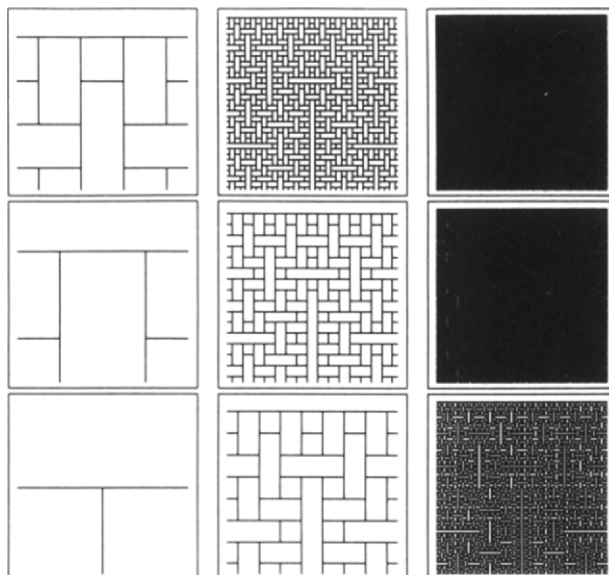
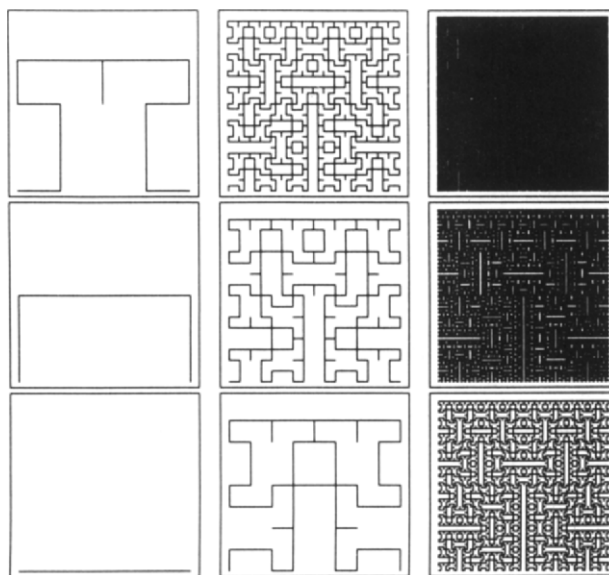


Figure VI.184. A sequence of curves “converging to” a space-filling curve. These are obtained by application of the Deterministic Algorithm to the IFS code in Table VI.1, starting from a curve A_0 , which connects $(0, 0)$ to $(1, 0)$ and lies in ■.



$$G_{yz} = \{(y, z) \in \mathbb{R}^2 : (x, y, z) \in G\},$$

namely the projection of G into the (y, z) plane. It is straightforward to prove that G_{yz} is the attractor $A = \blacksquare$ of the IFS defined by the IFS code in Table VI.1. It follows that $f([0, 1]) = \blacksquare$. So we have our space-filling curve!

We have something else very exciting as well. The attractor of the three-dimensional IFS is the graph of a function from $[0, 1]$ to $\blacksquare$. The projections G_{xy} and G_{xz} , in the obvious notation, are graphs of hidden-variable fractal functions, while $G_{yz} = \blacksquare$. What does G look like from other points of view? Various views of the attractor are illustrated in Figures VI.185 and VI.186. We conclude that G is a curious, complex, three-dimensional object. It would be wonderful to have a three-dimensional model of G made out of very thin strong wire.

The following theorem summarizes what we have just learned.

Theorem 5.1 *Let $A \subset \mathbb{R}^2$ be a nonempty pathwise-connected compact set, such that the following conditions hold. Let N be an integer greater than 1. Let there be a hyperbolic IFS $\{\mathbb{R}^2; M_n, n = 1, 2, \dots, N\}$ such that A is the attractor of the IFS. Let there be a set of distinct points $\{(F_i, G_i) \in A : i = 0, 1, 2, \dots, N\}$ such that*

$$M_n(F_0, H_0) = (F_{n-1}, H_{n-1}) \text{ and } w_n(F_N, H_N) = (F_n, H_n) \text{ for } n = 1, 2, \dots, N.$$

Then there is a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1]) = A$. One such function is the one whose graph is the attractor of the IFS

$$\{\mathbb{R}^3; w_n(x, y, z) = (\frac{1}{N}x + \frac{n-1}{N}, M_n(y, z)), n = 1, 2, \dots, N\}.$$

Examples & Exercises

5.1. Let Δ denote the Sierpinski triangle with vertices at the points $(0, 0)$, $(0, 1)$, and $(1, 0)$. Find an IFS of the form $\{\mathbb{R}^3; w_1, w_2, w_3\}$, where the maps are affine, such that the attractor of the IFS is the graph of a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1]) = \Delta$. Four projections of such an attractor are shown in Figure VI.187.

5.2. Find an IFS $\{\mathbb{R}^3; w_1, w_2, w_3, w_4\}$, where the transformations are affine, whose attractor is the graph of a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1]) = A$, where A is the set represented in Figure VI.188.

Figure VI.185. Various views of the attractor of a certain IFS. From some points of view we see that it is the graph of a function. From one point of view it is the graph of a space-filling curve!

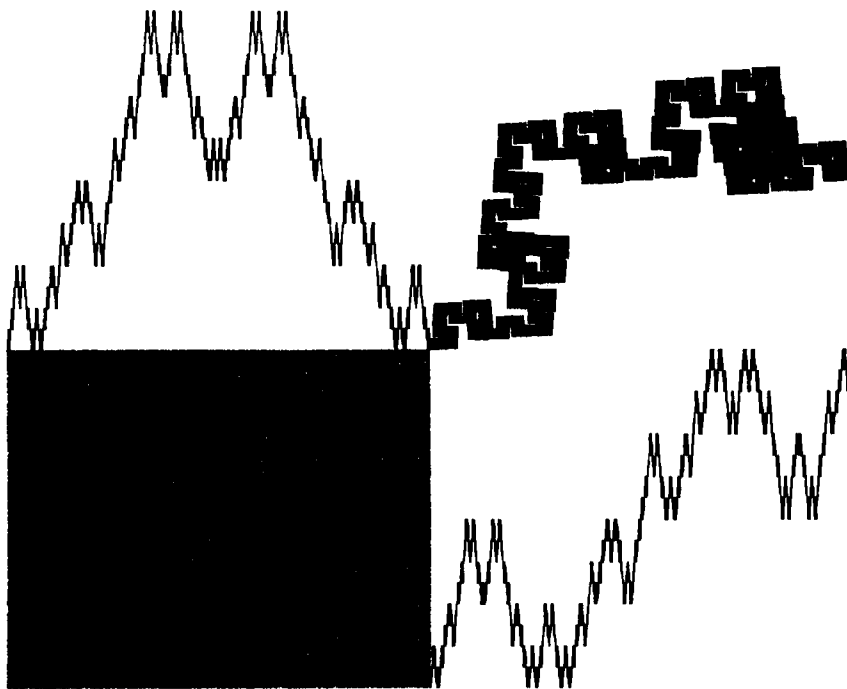
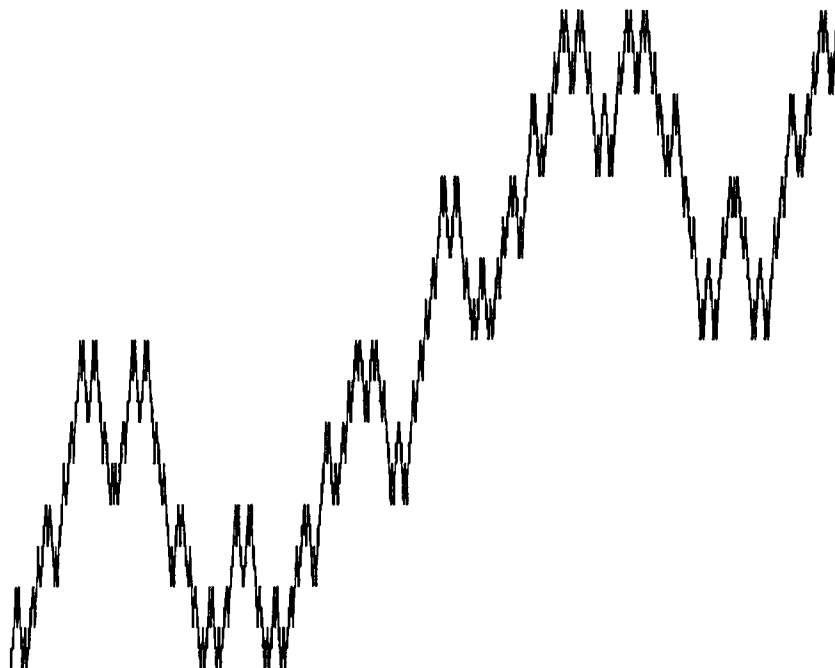


Figure VI.186. Higher-resolution view of the lower right panel of Figure VI.185.



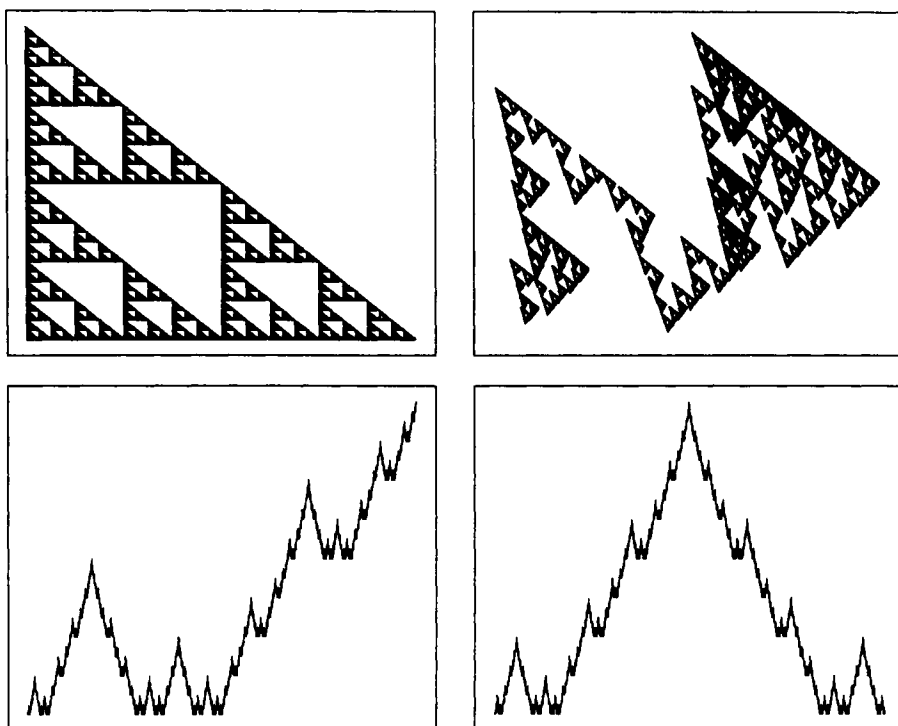


Figure VI.187. Four views of the attractor of an IFS. This attractor is the graph of a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1])$ is a Sierpinski triangle. This function provides a “space-filling” curve, where space is a fractal!

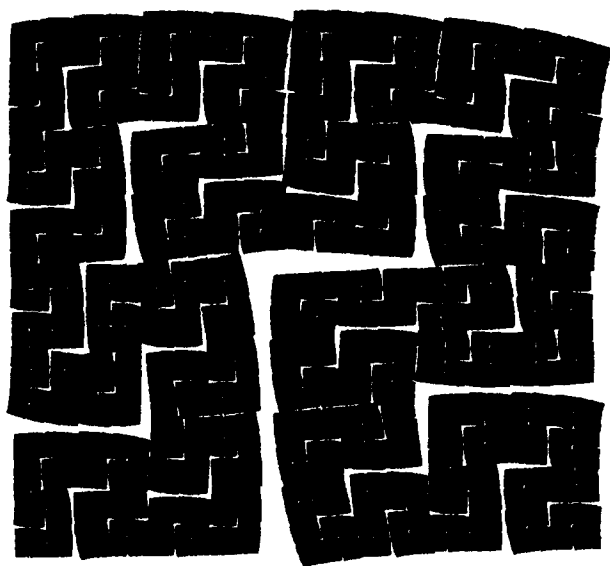


Figure VI.188. Find an IFS $\{\mathbb{R}^3; w_1, w_2, w_3, w_4\}$, where the transformations are affine, whose attractor is the graph of a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$ such that $f([0, 1]) = A$, where A is the set represented here.